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(54) TRUSS ASSEMBLY TABLE

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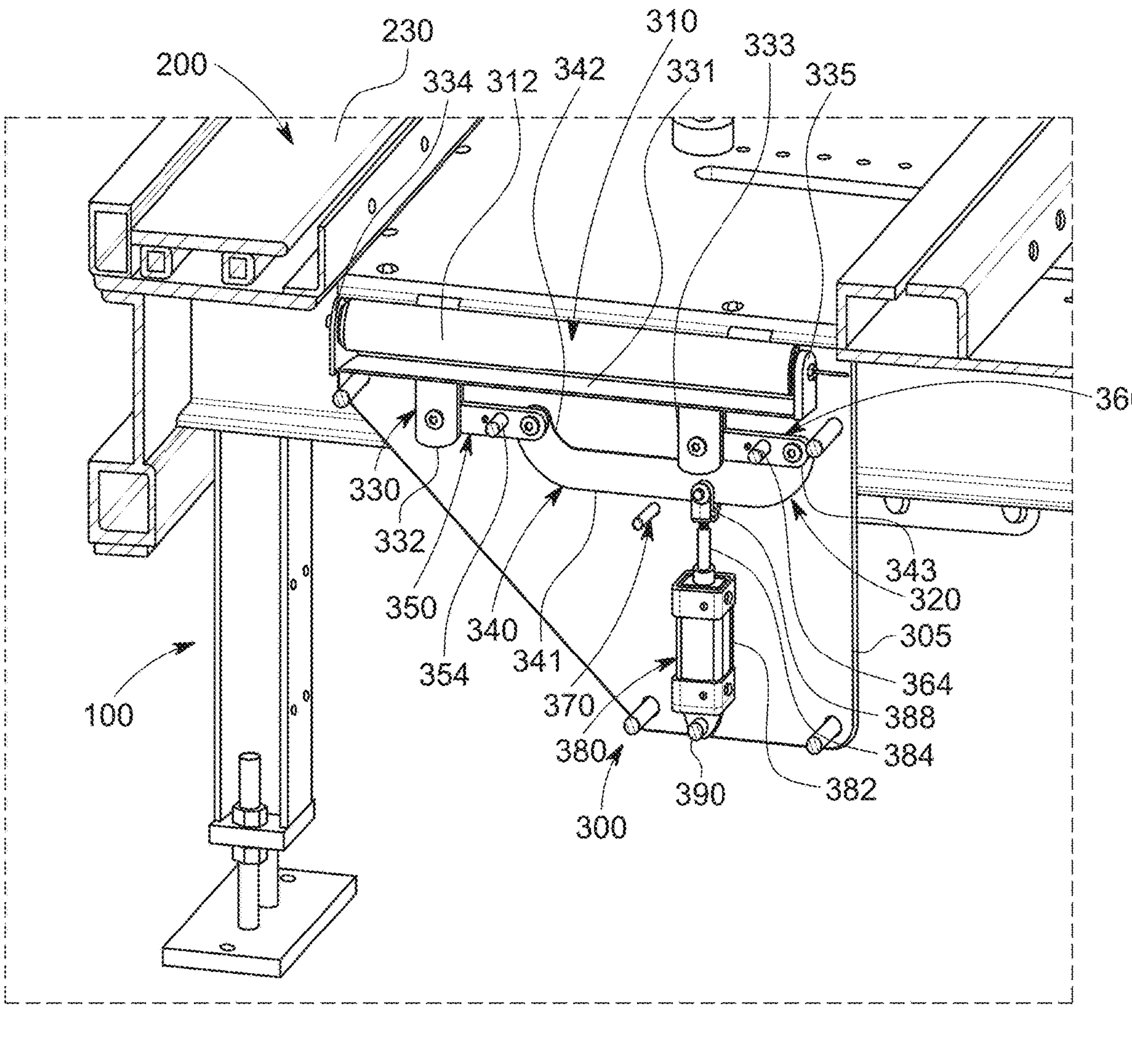
(57) **ABSTRACT**

A truss assembly table and methods of operating a truss assembly table, wherein the truss assembly table includes one or more of truss ejection assemblies, truss end stop assemblies, and a control system and gantry with a sensor assembly.



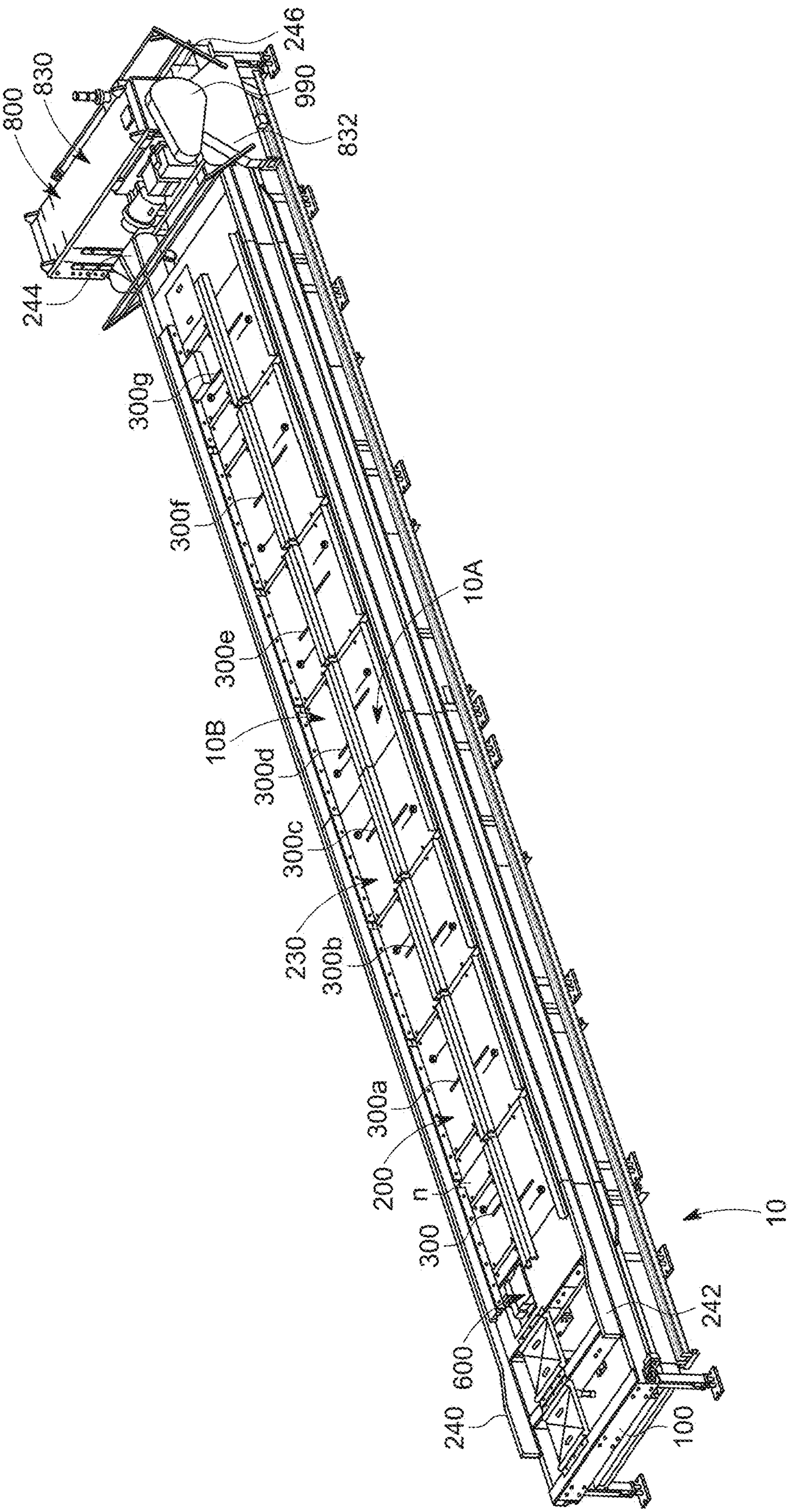

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FIG. 1

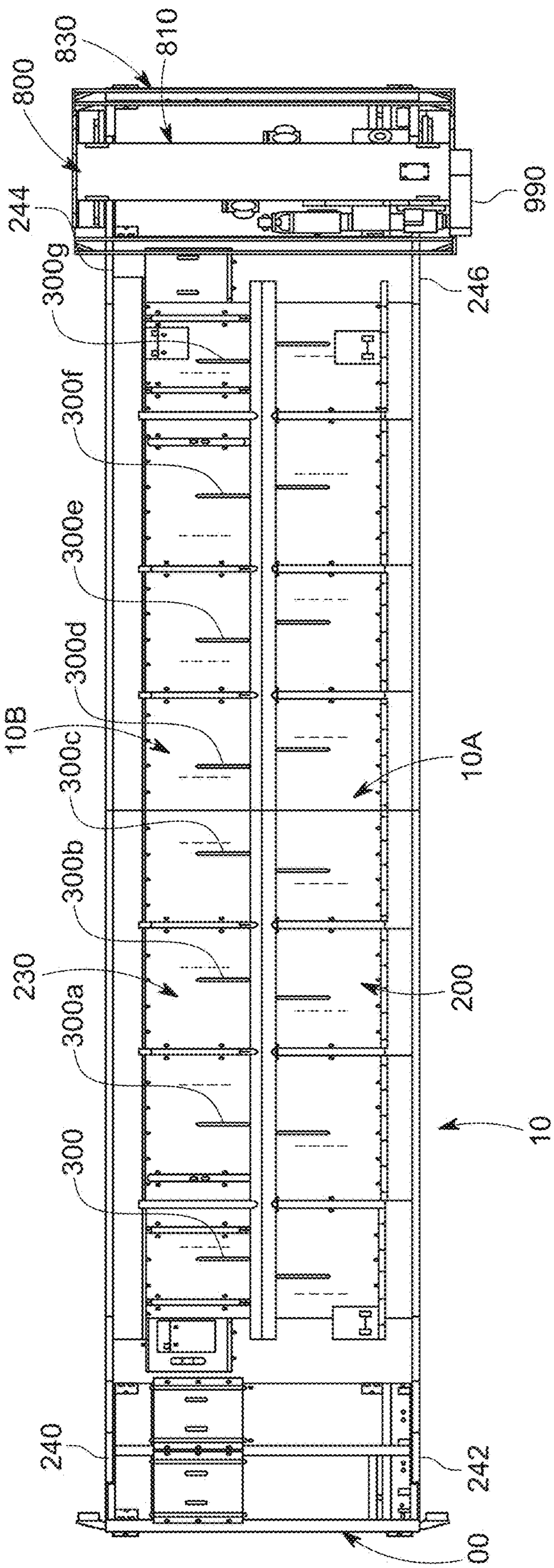
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FIG. 2

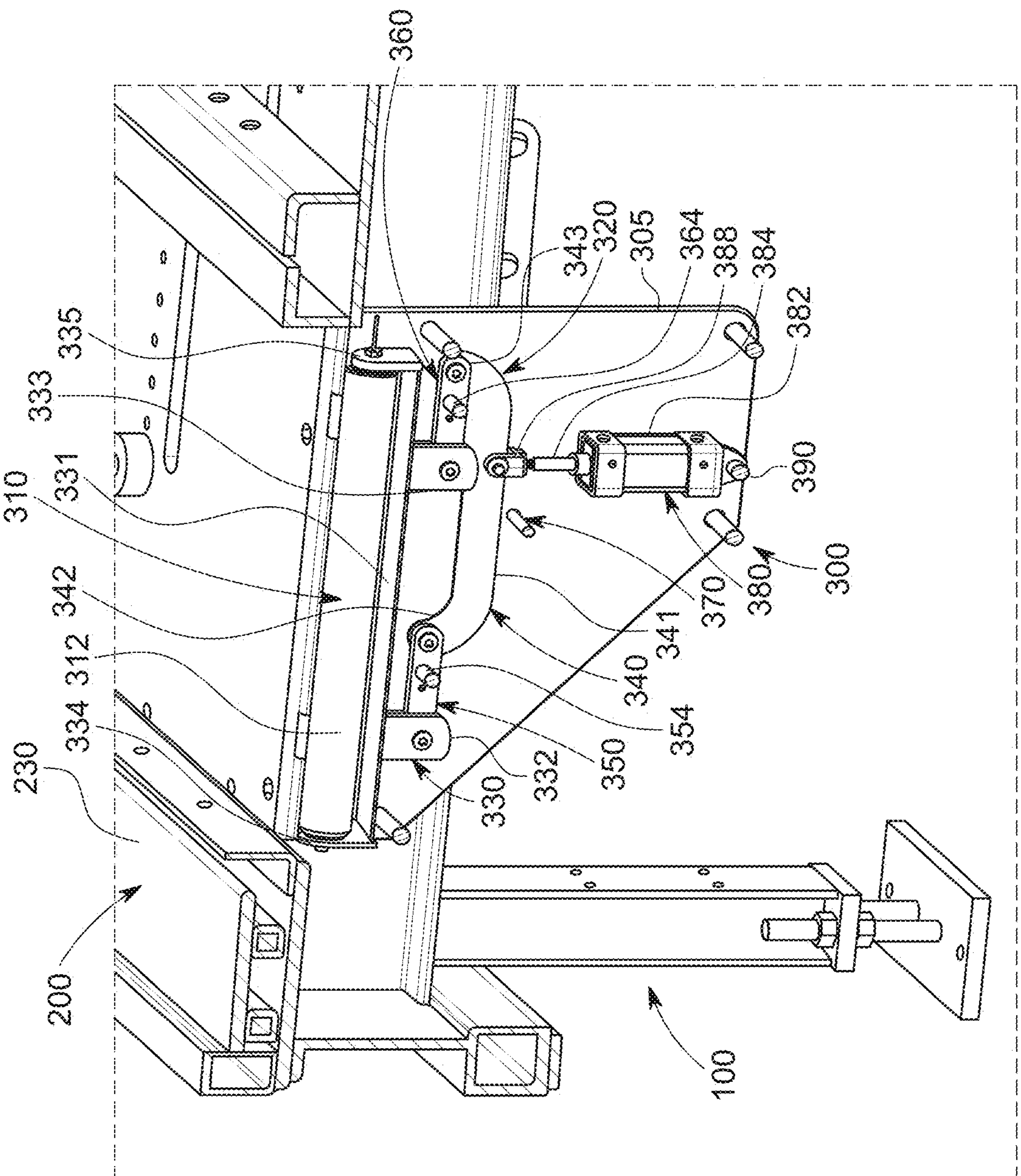
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FIG. 3

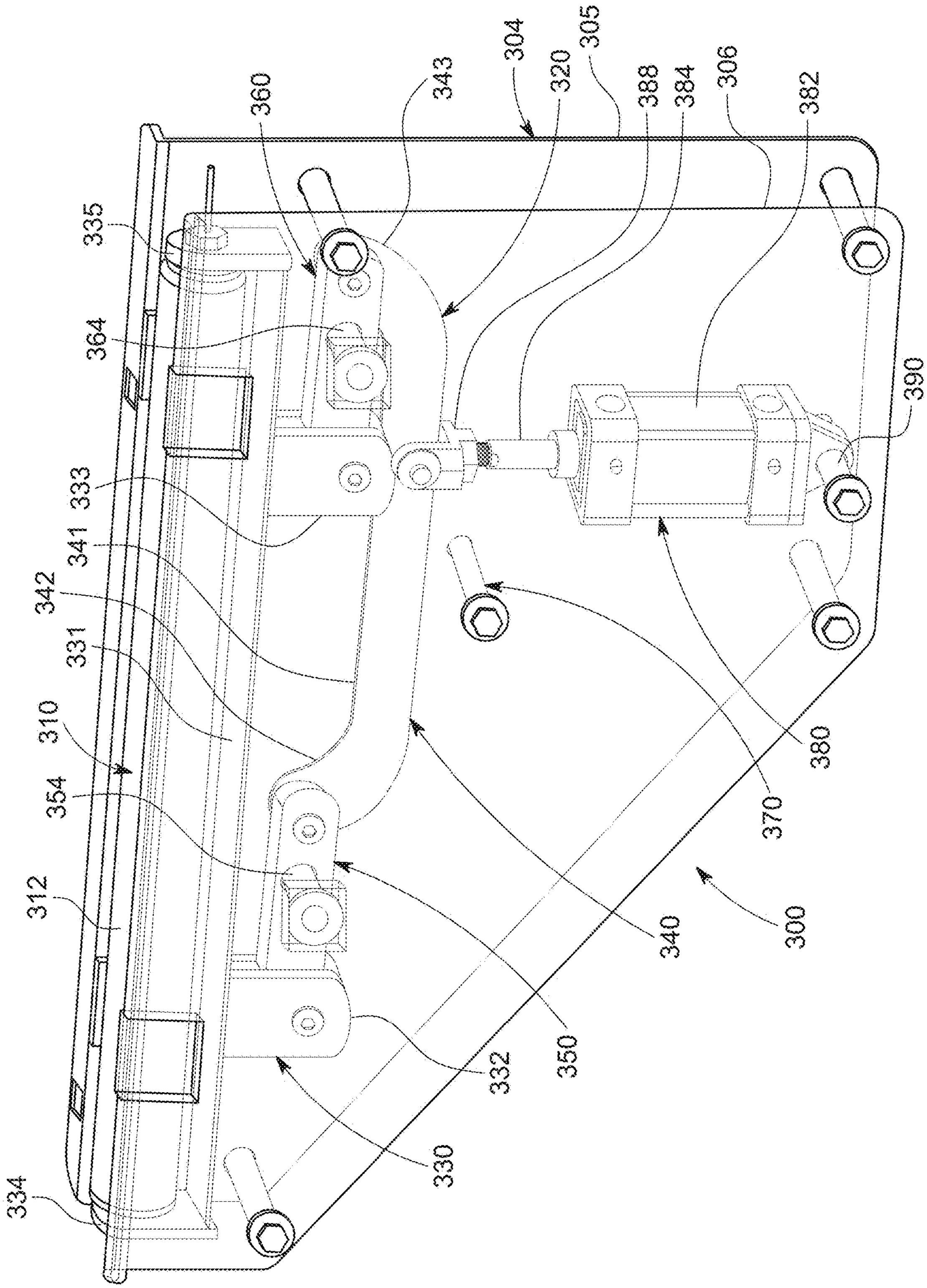
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FIG. 4

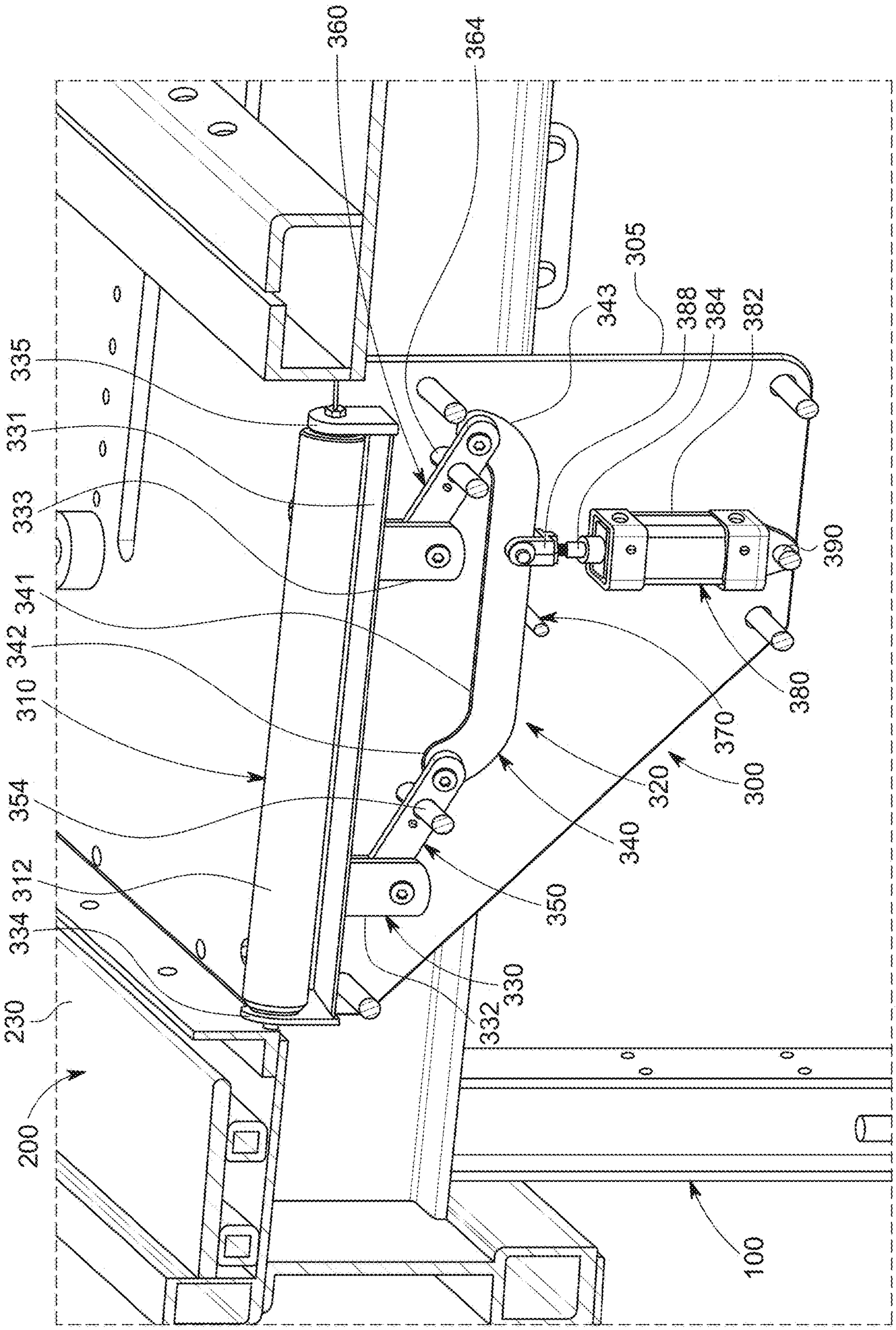
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FIG. 5

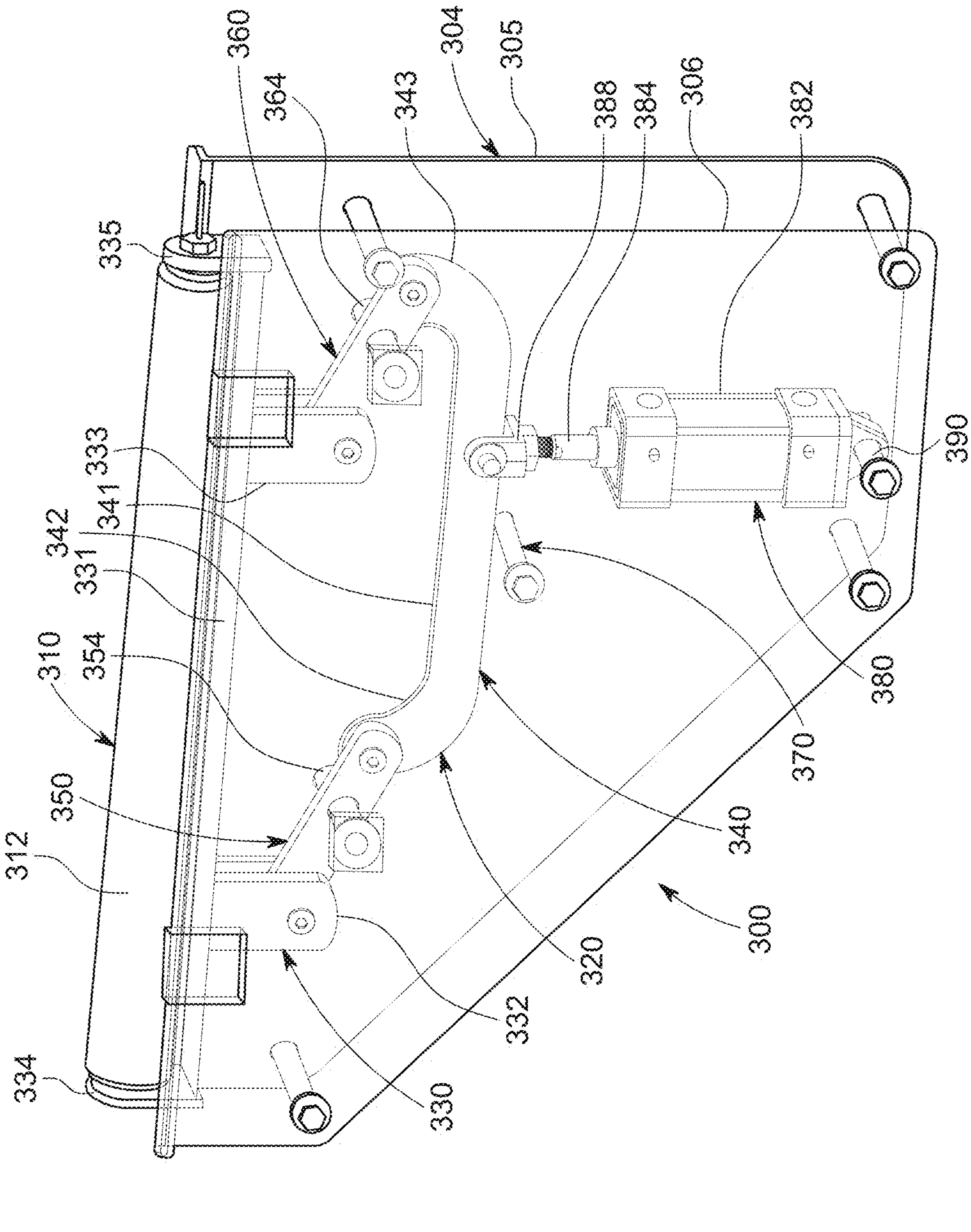
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FIG. 6

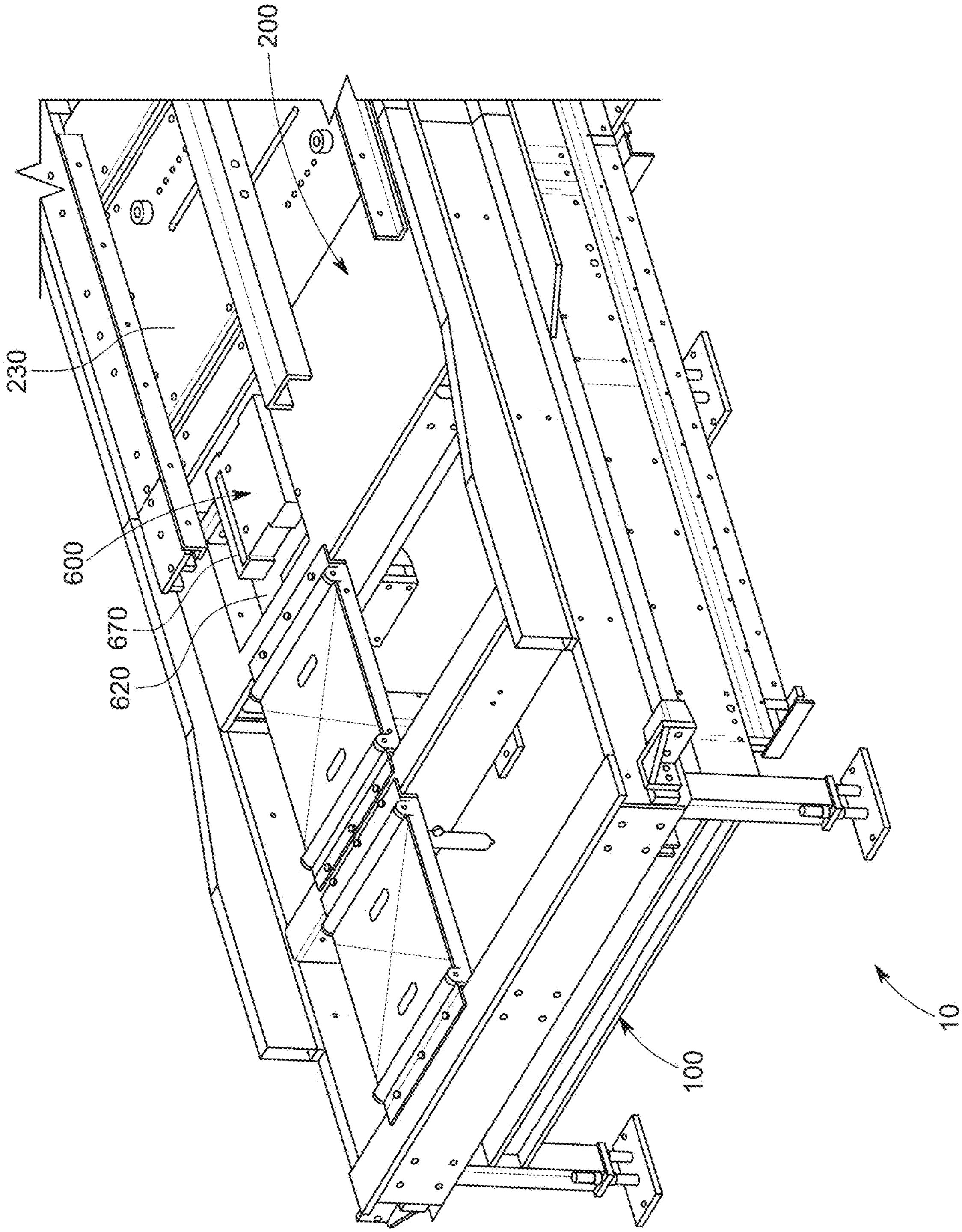
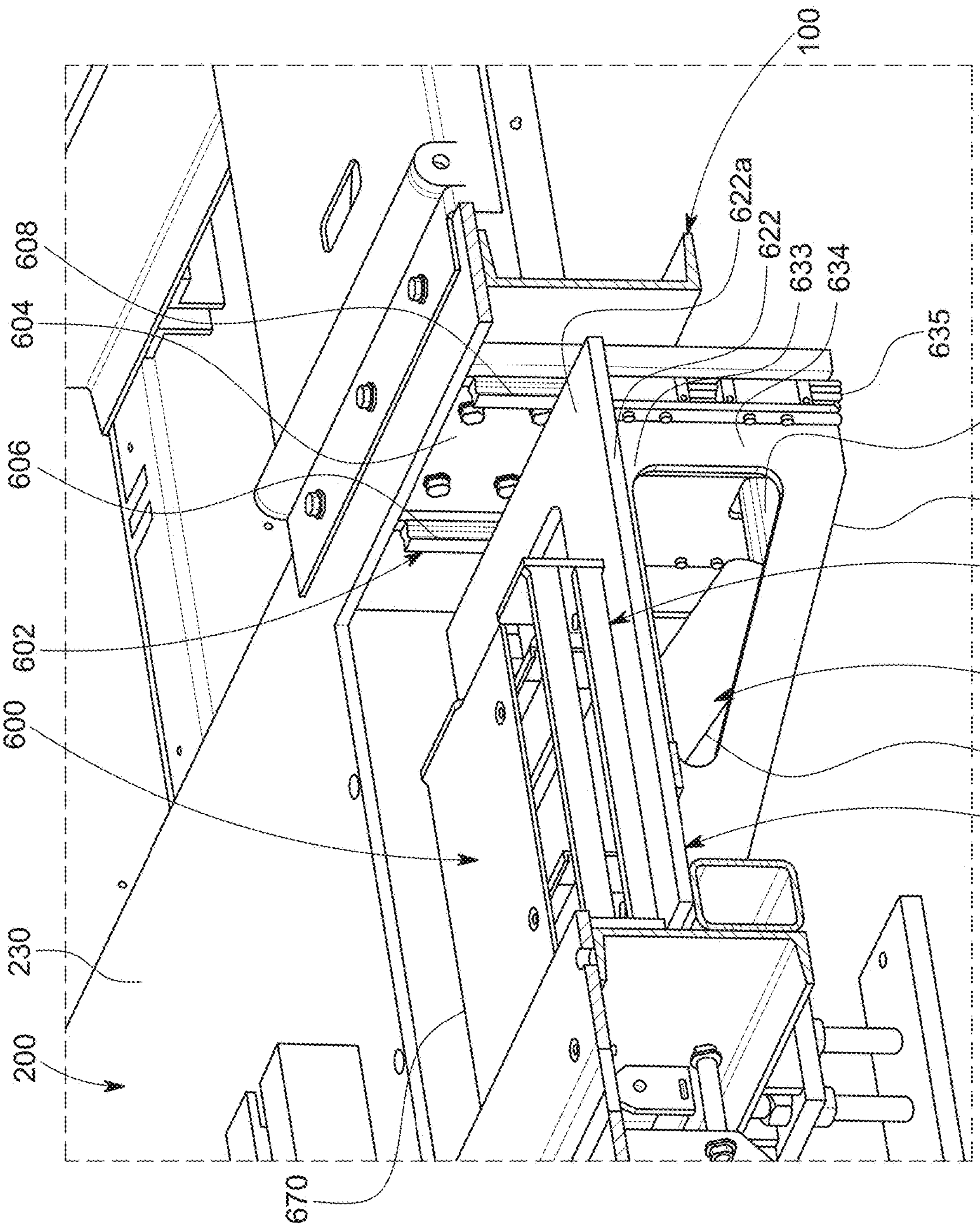


FIG. 7


100
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230
200
670

FIG. 8

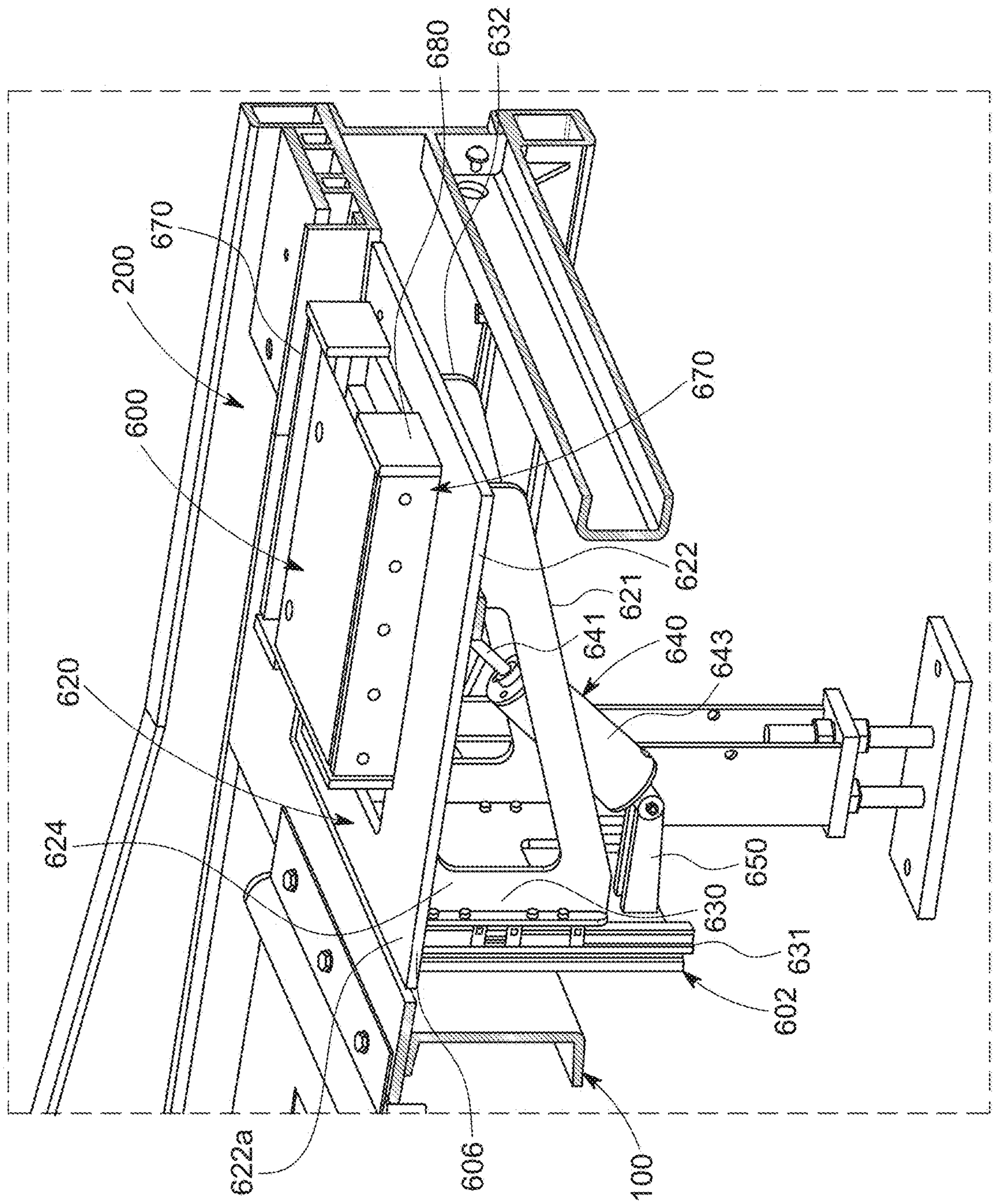
680
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FIG. 9

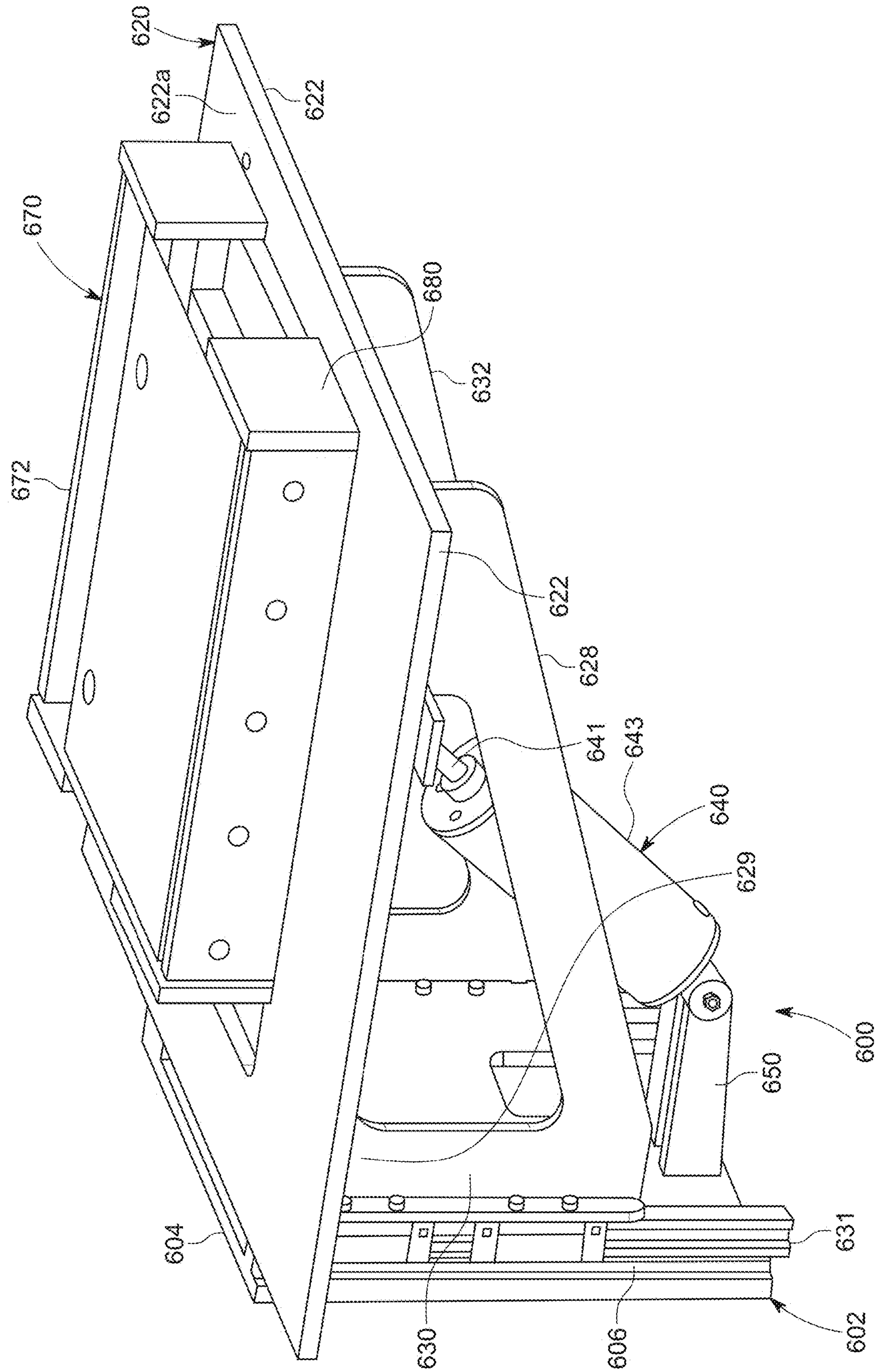
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FIG. 10

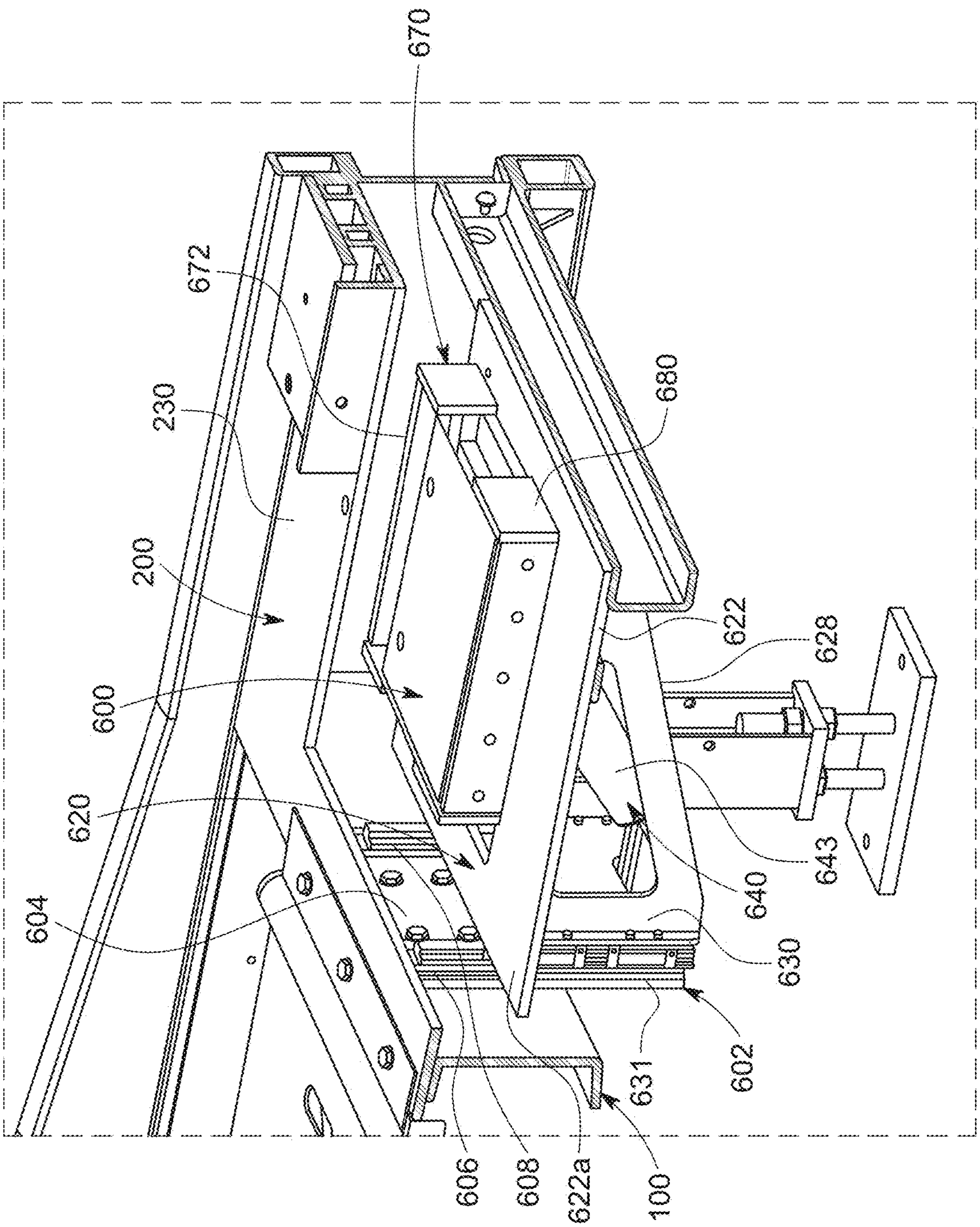
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FIG. 11

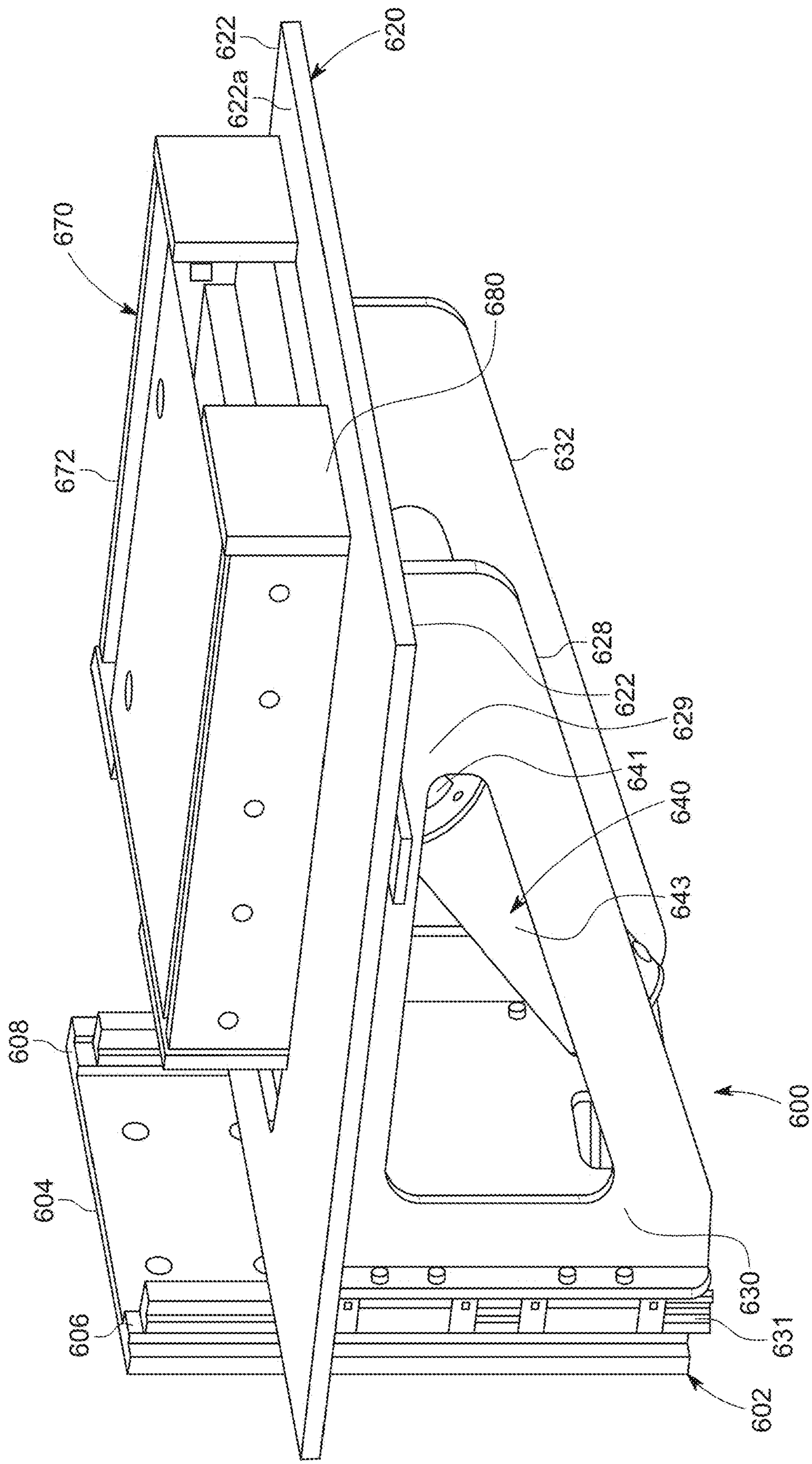
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FIG. 12

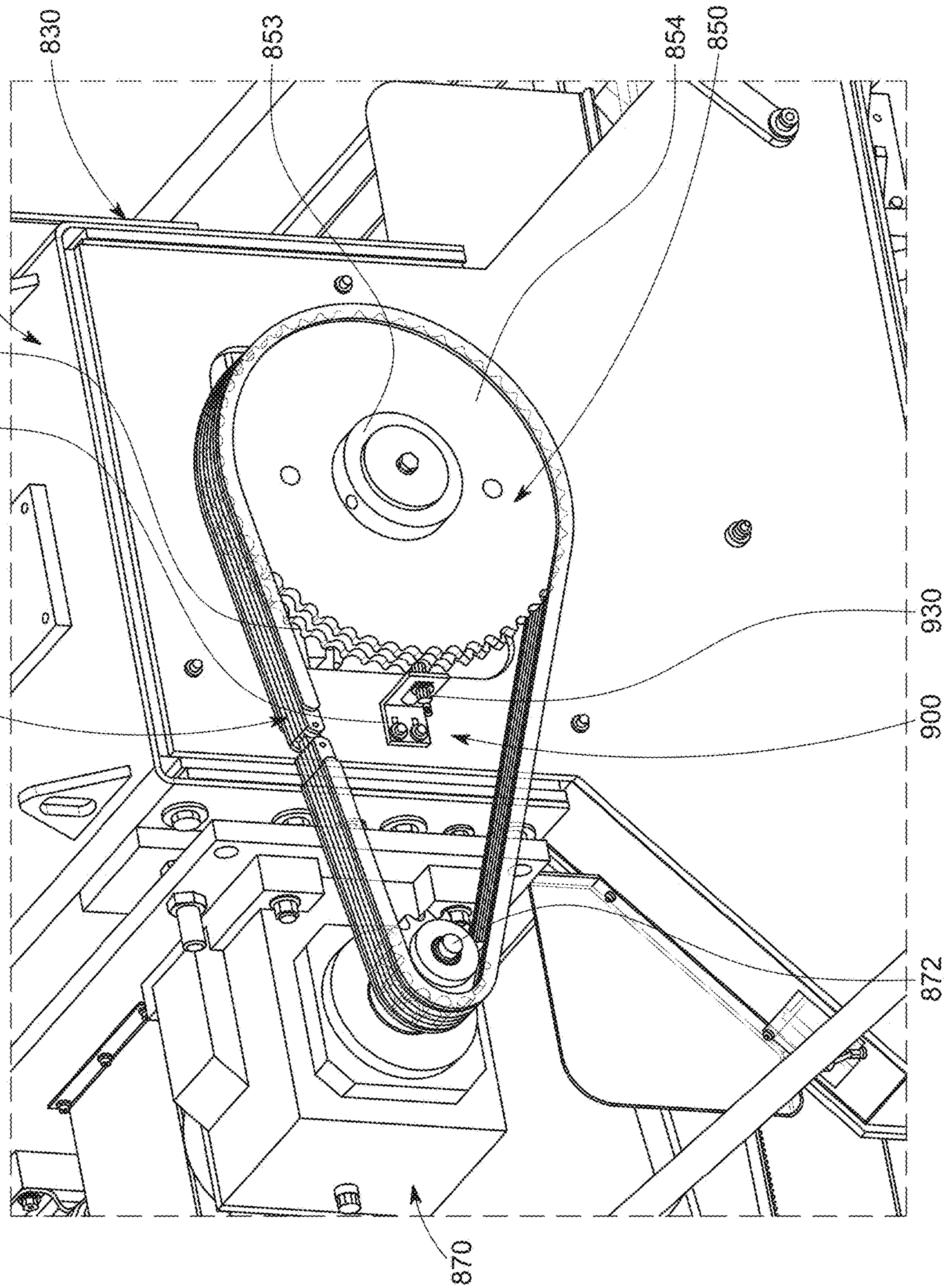

FIG. 13

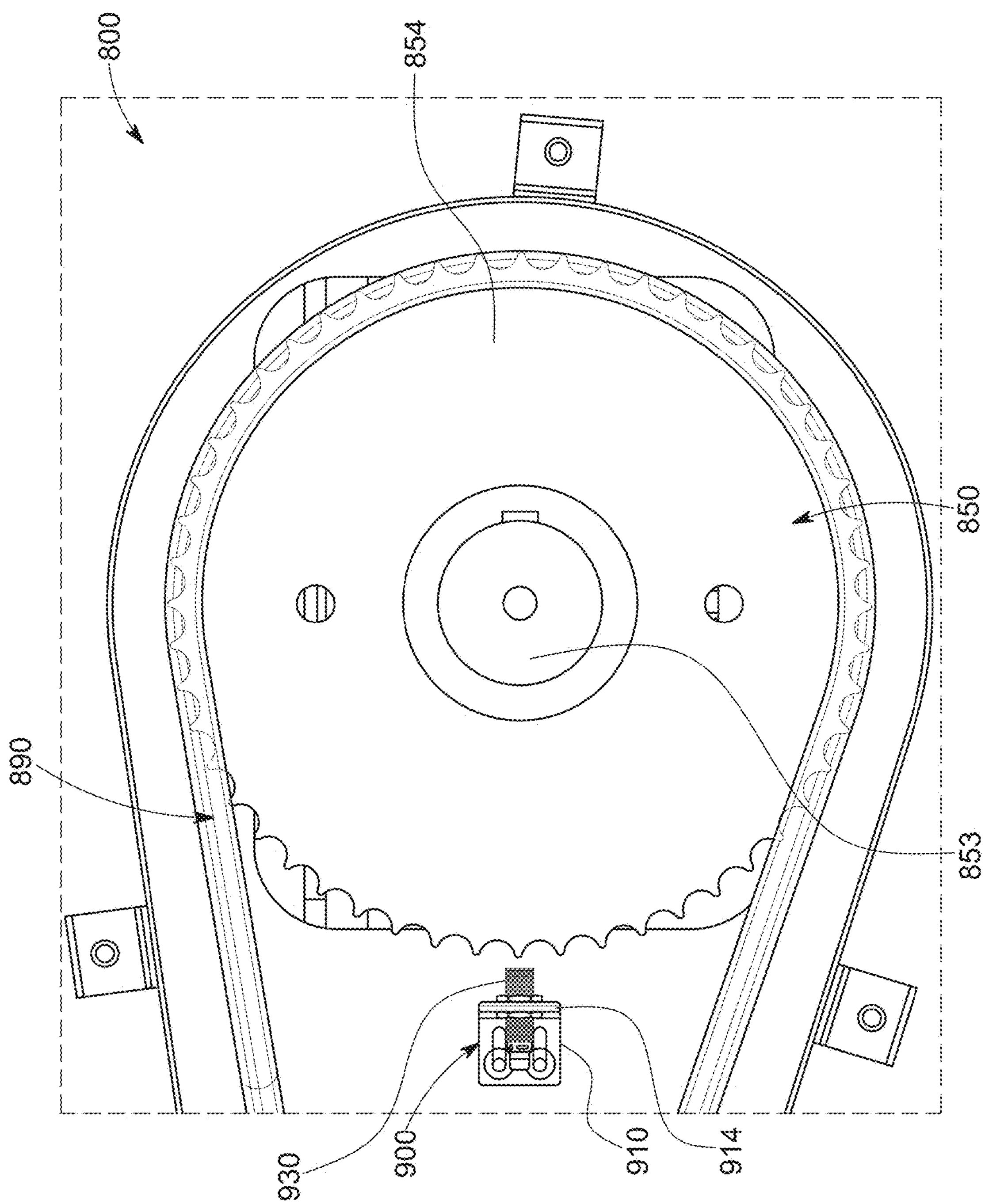

FIG. 14

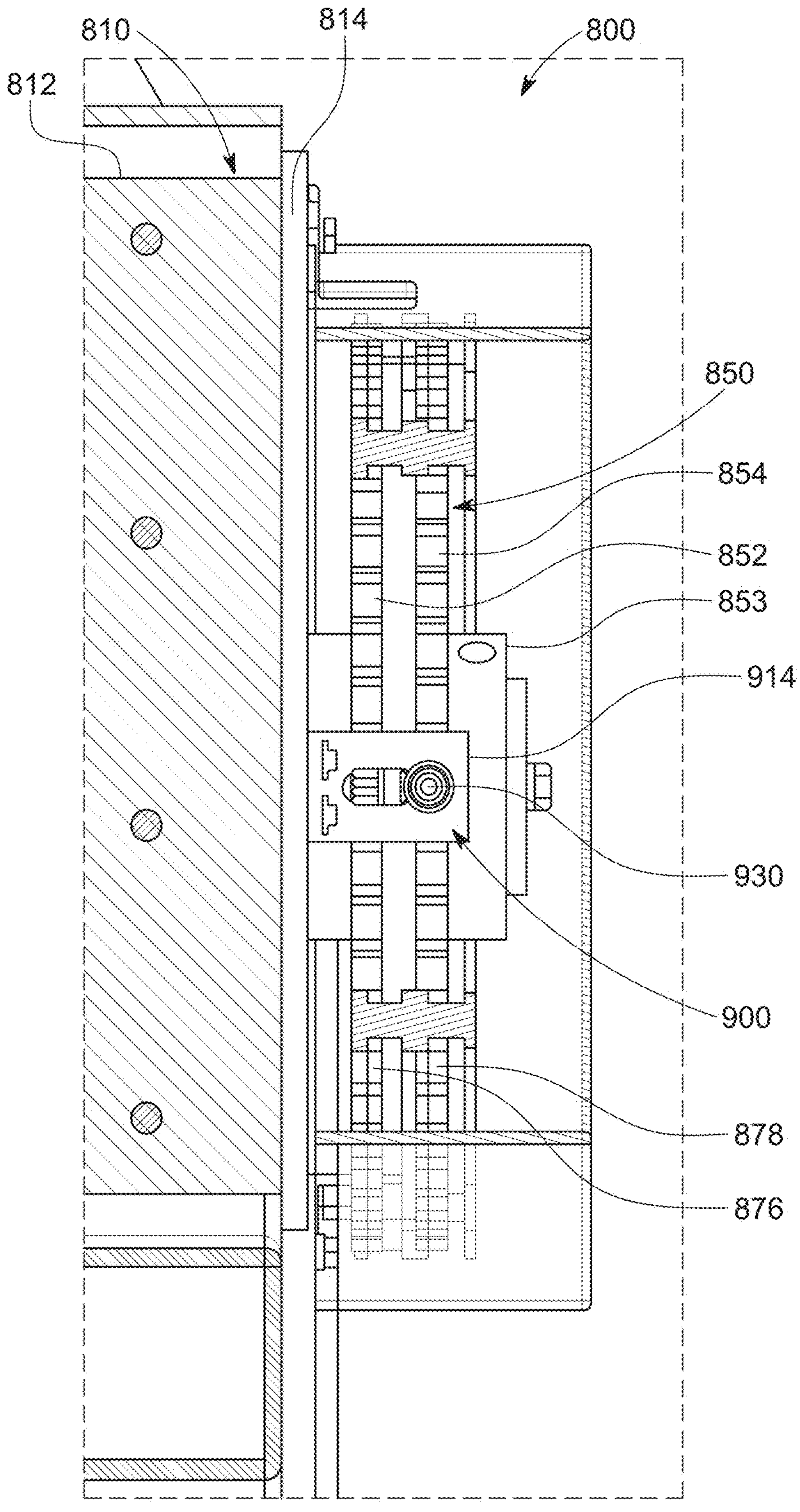

FIG. 15

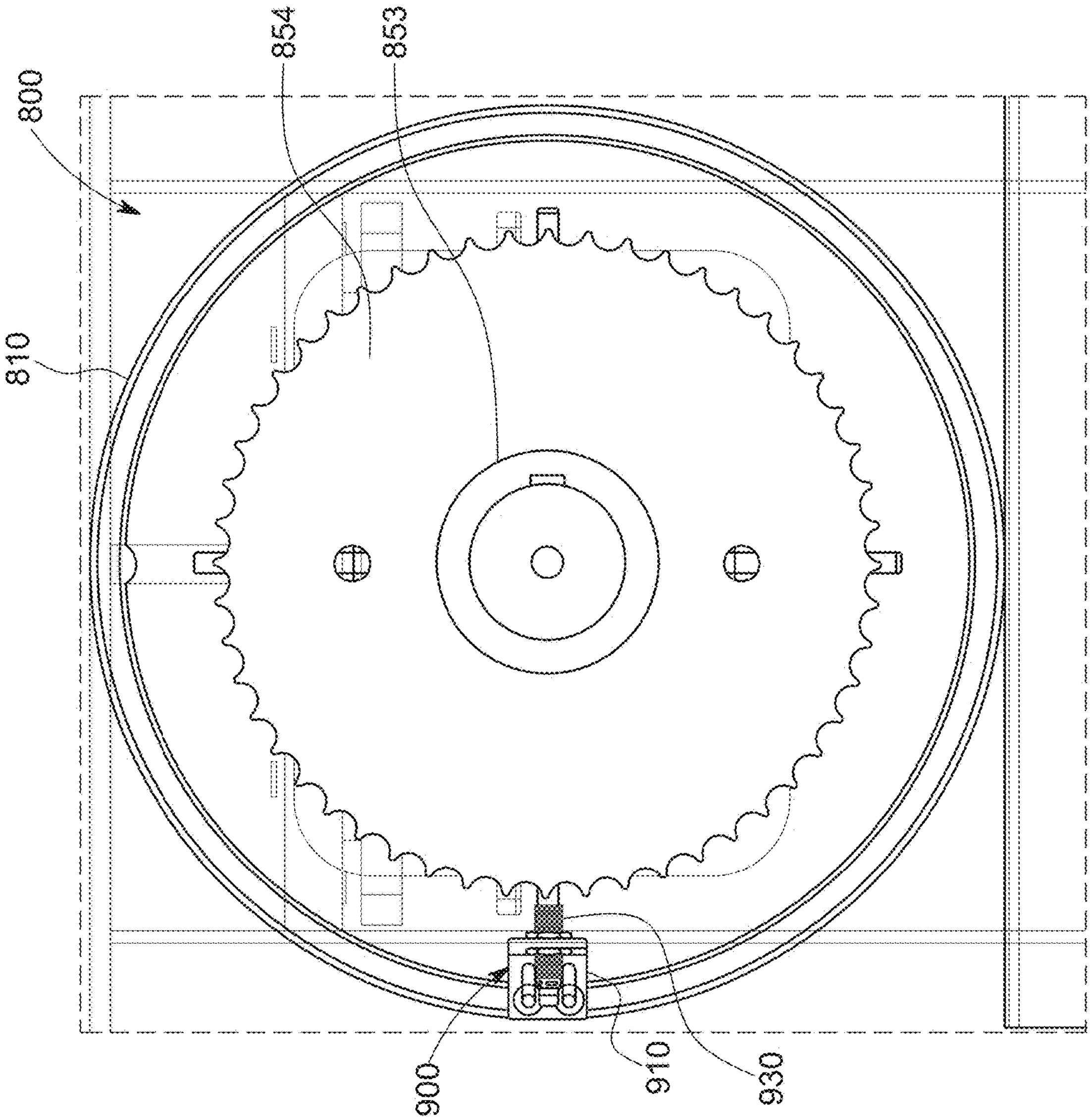

FIG. 16

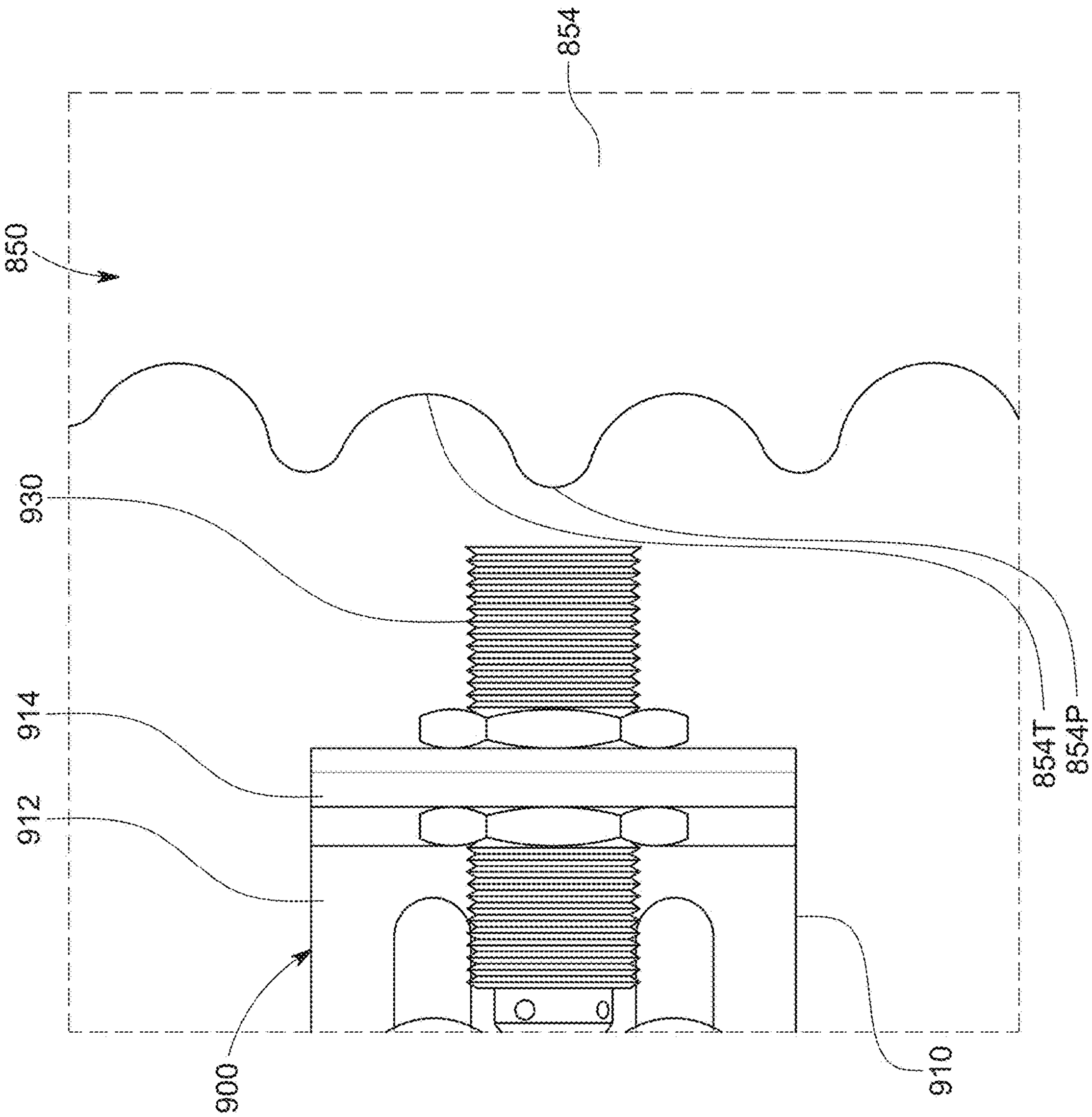

FIG. 17

TRUSS ASSEMBLY TABLE

PRIORITY CLAIM

[0001] This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/562,403, filed Mar. 7, 2024, the entire contents of which are incorporated hereon by reference.

BACKGROUND

[0002] Wooden trusses are widely used throughout the construction industry. Wooden trusses are often built from conventional dimensional lumber members (such as what is commonly known as: a 2 by 4; a 2 by 6; a 2 by 8; etc.). The wooden members that are used to build a wooden truss are sometimes called truss members in general with the most common truss member types sometimes called chord members and web members. Such chord members extend longitudinally along the length of the truss and such web members extend transversely to the length of the truss such as along the width of the truss. A wooden truss is often built from numerous wooden truss members and metal connectors. The metal connectors are used to attach the truss members to build the wooden truss. Wooden trusses are often prefabricated in a factory and then shipped to a construction site where the wooden trusses are used to construct part of the structure of a building (such as a house or commercial facility). Buildings constructed with such prefabricated wooden roof trusses are often more economical and faster to construct than buildings constructed with conventional stick framed structures.

[0003] Various truss assembly tables have been developed and commercialized.

[0004] One issue with various known truss assembly tables is that they are only configured to eject the built trusses from the side of the truss assembly tables. In other words, such various known truss assembly tables are not configured to eject the built trusses from either or both of the ends of the truss assembly tables.

[0005] Another issue with various known truss assembly tables is that they are not configured to automatically control the movement of the gantry of the truss assembly table that fully secures the connector plates to the truss members such as the chord members and web members.

SUMMARY

[0006] The present disclosure provides truss assembly tables, methods of operation thereof, and various components and combinations of components thereof.

[0007] In various embodiments, the present disclosure relates to a truss assembly table including a plurality of truss ejection assemblies configured to simultaneously be in: (1) retracted positions where the truss ejection assemblies are at or below the top surface of the table top of the truss assembly table; and (2) extended positions wherein rollers of the truss ejections assemblies raise above the top surface of the table top of the truss assembly table and engage, raise, and eject a truss built on the truss assembly table from an end of the truss assembly table. In various embodiments, the present disclosure also relates to each such truss ejection assembly configured to installed as part of a truss assembly table. In various embodiments, the present disclosure also relates to methods of operating such truss ejection assemblies.

[0008] In various embodiments, the present disclosure relates to a truss assembly table including a truss end stop assembly configured to be in: (1) an engagement position where the truss end stop assembly is configured to engage one or more end truss members of the truss during building of the truss on the truss assembly table; and (2) a non-engagement position where the truss end stop assembly is disengaged from the end truss member(s) of the built truss and enables the built truss to move above the truss end stop assembly as the built truss is ejected from an end of the truss assembly table. In various embodiments, the present disclosure also relates to such a truss end stop assembly configured to installed as part of a truss assembly table. In various embodiments, the present disclosure also relates to methods of operating such truss end stop assembly or assemblies.

[0009] In various embodiments, the present disclosure relates to a truss assembly table including a control system and a gantry including a sensor assembly configured to co-act to sense movement of a gear component of the gantry drive assembly and to determine distances of movement of the gantry and/or to control the distances of movement of the gantry in along the table top (such as in forward and backward directions). In various embodiments, present disclosure also relates to such a control system and sensor assembly configured to installed as part of a truss assembly table. In various embodiments, the present disclosure also relates to methods of operating such a control system and gantry including a sensor assembly.

[0010] In various embodiments, the present disclosure relates to a truss assembly table including: (1) such truss ejection assemblies; and (2) such truss end stop assembly. In various embodiments, the present disclosure also relates to methods of operating such components.

[0011] In various embodiments, the present disclosure relates to a truss assembly table including: (1) such truss ejection assemblies; (2) such truss end stop assembly; and (3) such control system and gantry with a sensor assembly. In various embodiments, the present disclosure also relates to methods of operating such components.

[0012] In various embodiments, the present disclosure relates to a truss assembly table including: (1) such truss ejection assemblies; and (2) such control system and gantry with a sensor assembly. In various embodiments, the present disclosure also relates to methods of operating such components.

[0013] In various embodiments, present disclosure relates to a truss assembly table including: (1) such truss end stop assembly; and (2) such control system and gantry including a sensor assembly. In various embodiments, the present disclosure also relates to methods of operating such components.

[0014] In various embodiments, present disclosure also relates to a truss assembly table including a table frame and/or table top that includes an upwardly extending set of spaced apart ramps at each end thereof. The gantry is configured to travel upwardly on each set of ramps and thus be spaced a further distance from the top of the table top. This better enables a built truss to move under the gantry as the built truss is ejected from an end of the truss assembly table.

[0015] Additional features and advantages of the present disclosure are described in, and will be apparent from, the following Detailed Description and the Figures.

BRIEF DESCRIPTION OF THE FIGURES

[0016] FIG. **1** is a top perspective view of a truss assembly table in accordance with one example embodiment of the present disclosure.

[0017] FIG. **2** is a top view of the truss assembly table of FIG. **1**.

[0018] FIG. **3** is an enlarged fragmentary partial cross-sectional and partial perspective view of part of the truss assembly table of FIG. **1**, showing one of the truss ejection assemblies thereof in a retracted position.

[0019] FIG. **4** is an enlarged perspective view of the truss ejection assembly of FIG. **3** shown removed from the truss assembly table, and wherein the truss ejection assembly is in the retracted position.

[0020] FIG. **5** is an enlarged fragmentary partial cross-sectional and partial perspective view of part of the truss ejection assembly of FIG. **1**, and wherein the truss ejection assembly is in an extended position.

[0021] FIG. **6** is an enlarged perspective view of the truss ejection assembly of FIG. **3** shown removed from the truss assembly table, and wherein the truss ejection assembly is in the extended position.

[0022] FIG. **7** is an enlarged fragmentary perspective view of part of the truss assembly table of FIG. **1**, showing the truss end stop assembly at one end to the truss assembly table, and wherein the truss end stop assembly is in a lower non-engagement position.

[0023] FIG. **8** is a further enlarged fragmentary partial cross-sectional and partial perspective view of part of the truss assembly table of FIG. **1**, showing the truss end stop assembly at one end to the truss assembly table, and wherein the truss end stop assembly is in a lower non-engagement position.

[0024] FIG. **9** is a further enlarged fragmentary partial cross-sectional and partial perspective view of part of the truss assembly table of FIG. **1**, showing the truss end stop assembly at one end to the truss assembly table, and wherein the truss end stop assembly is in an upper engagement position.

[0025] FIG. **10** is a further enlarged perspective view of the truss end stop assembly of FIG. **7** shown removed from the truss assembly table, and wherein the truss end stop assembly is in the upper engagement position.

[0026] FIG. **11** is a further enlarged fragmentary partial cross-sectional and partial perspective view of part of the truss assembly table of FIG. **1**, showing the truss end stop assembly at one end to the truss assembly table, and wherein the truss end stop assembly is in a lower non-engagement position.

[0027] FIG. **12** is a further enlarged perspective view of the truss end stop assembly of FIG. **7** shown removed from the truss assembly table, and wherein the truss end stop assembly is in the lower non-engagement position.

[0028] FIG. **13** is an enlarged fragmentary perspective view of part of the truss assembly table of FIG. **1** showing part of the gantry thereof with a side cover of the gantry removed.

[0029] FIG. **14** is an enlarged fragmentary side view of part of the gantry of the truss assembly table of FIG. **1** with the side cover of the gantry removed.

[0030] FIG. **15** is an enlarged fragmentary partial cross-sectional front-end view of part of the gantry of the truss assembly table of FIG. **1** with the side cover of the gantry removed.

[0031] FIG. **16** is a further enlarged fragmentary side view of part of the gantry of the truss assembly table of FIG. **1** with the side cover of the gantry removed.

[0032] FIG. **17** is a still further enlarged fragmentary side view of part of the gantry of the truss assembly table of FIG. **1** with the side cover of the gantry removed.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0033] While the systems, devices, and methods described herein may be embodied in various forms, the drawings show, and the specification describes certain exemplary and non-limiting embodiments. Not all components shown in the drawings and described in the specification may be required, and certain implementations may include additional, different, or fewer components. Variations in the arrangement and type of the components; the shapes, sizes, and materials of the components; and the manners of connections of the components may be made without departing from the spirit or scope of the claims. Unless otherwise indicated, any directions referred to in the specification reflect the orientations of the components shown in the corresponding drawings and do not limit the scope of the present disclosure. Further, terms that refer to mounting methods, such as mounted, connected, etc., are not intended to be limited to direct mounting methods but should be interpreted broadly to include indirect and operably mounted, connected, and like mounting methods. This specification is intended to be taken as a whole and interpreted in accordance with the principles of the present disclosure and as understood by one of ordinary skill in the art.

[0034] Various embodiments of the present disclosure relate to truss assembly tables and particularly provide a truss assembly table with: (1) multiple truss ejection assemblies configured to eject built trusses from an end of the truss assembly table; (2) one or more truss end stop assemblies configured to secure one or more truss member(s) of the truss while the truss is being built on the truss assembly table and to move downwardly to enable the built truss to move over the truss end stop assembly and to be ejected from an end of the truss assembly table; and (3) a control system and gantry including a sensor assembly. Thus, in various embodiments, the present disclosure relates to truss assembly tables with all three of: (1) the truss ejection assemblies; (2) the truss end stop assembly; and (3) the control system and gantry with the sensor assembly. Various embodiments of the present disclosure also relate to methods of operating a truss assembly table with such components.

[0035] In various embodiments, the present disclosure relates to truss assembly tables with two of but not all of: (1) the truss ejection assemblies; (2) the truss end stop assembly; and (3) the control system and gantry with the sensor assembly. Various embodiments of the present disclosure also relate to methods of operating a truss assembly table with such components.

[0036] In various embodiments, the present disclosure relates to truss assembly tables with only one of: (1) the truss ejection assemblies; (2) the truss end stop assembly; and (3) the control system and gantry with the sensor assembly. Various embodiments of the present disclosure also relate to methods of operating a truss assembly table with such components.

[0037] In various embodiments, the present disclosure relates individually to each of: (1) the truss ejection assemblies; (2) the truss end stop assembly; and (3) the control system and gantry with the sensor assembly.

[0038] In various embodiments, the present disclosure also relates to methods of operation (jointly or separately) of each of: (1) the truss ejection assemblies; (2) the truss end stop assembly; and (3) the control system and gantry with sensor assembly, on a truss assembly table, and/or the operation of the truss assembly table with one or more or all of these components.

[0039] In various embodiments, present disclosure also relates to a truss assembly table including a table frame and/or table top that includes an upwardly extending set of spaced apart ramps at one or both ends of the table frame and/or table top. In such embodiments, the gantry is configured to travel upwardly on each set of ramps and thus be spaced a further distance from the top surface of the table top. This better enables a built truss to move under the gantry as the built truss is ejected from an end of the table.

[0040] Example embodiments of the truss assembly table including these components of the present disclosure are discussed below; however, it should be appreciated that the present disclosure is not limited to the illustrated example truss assembly tables or the example components.

[0041] FIGS. **1** to **17** show a truss assembly table **10** (and various components thereof) of one example embodiment of the present disclosure. This example truss assembly table **10** generally includes: (a) a table frame **100**; (2) a table top **200**; (3) a plurality of spaced-apart ejection assemblies **300**, **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, and **300***f*; (4) a truss end stop assembly **600**; (5) a movable gantry **800**; (6) a control system (not shown); (7) an operator interface (not shown); and (8) a power supply (not shown).

[0042] For brevity, various components of this example truss assembly table **10** of the present disclosure are not described herein because such components are well known in the truss assembly table industry.

[0043] This example truss assembly table **10** is configured for simultaneously building two trusses (such as two floor trusses) on the table top **200** of the truss assembly table **10** (i.e., one on each side in and end-to-end manner on the table top **200**). Generally, to build a truss on the table top **200** of the truss assembly table **10**: (1) the truss members (not shown) and first side connectors (not shown) for a truss are first positioned on the first side **10**A of the truss assembly table **10** and the truss members are held in place by puck assemblies (not shown in detail or labeled); (2) the gantry **800** moves from a first end of to a second end of the table top **200** over the truss members and first side connectors to fully secure those first side connectors to the truss members; (3) the partially build truss is then flipped over from the first side **10**A to the second side **10**B of the truss assembly table **10**; and (4) the gantry **800** then moves from the second end back to the first end of the table top **200** over the truss member and second side connectors to fully secure second side connectors to the truss members and thus to complete the building of the truss. The built truss is then removed from an end of the second side **10**B of the truss assembly table **10**. The process of removing the truss includes lowering the end stop assembly (described below) and raising the truss ejection assemblies (described below) to lift the built truss on the second side **10**B of the truss assembly table **10**, and then causing the truss ejections assemblies to move the built truss longitudinally over the lowered end stop assembly and off of an end of the truss assembly table **10**. Thus, a first truss can be built on the first side **10**A while a second truss is built on the second side **10**B at the same time (although the stages of the building process for the first and second trusses will be different).

[0044] More specifically, the table frame **100** includes a plurality of frame components (not individually labeled) that are configured and attached in a suitable manner to support the other components of the truss assembly table **10**. These frame components can take any suitable shape, can be formed is any suitable manner, and can be formed from any suitable materials. The table frame **100** has a longitudinal length that is longer than any truss that will be built on the truss assembly table **10**. The table frame **100** also has a transverse width that is more than twice the width of any truss that will be built on the truss assembly table **10**, so as to enable two trusses (such as but not limited to floor trusses) to be simultaneously built on the truss assembly table **10**.

[0045] The table top **200** is supported by the table frame **100**, has a longitudinal length that is longer than any truss that will be built on the truss assembly table **10**, and has a transverse width that is more than twice the width of any truss that will be built on the truss assembly table **10**, so as to enable two trusses (such as but not limited to floor trusses) to be simultaneously built on the truss assembly table **10** as explained above. The table top **200** includes one or more horizontally extending members (not labeled), wherein the upper most member(s) form an upper build surface **230** on which trusses being built on the truss assembly table **10** can rest. The table top **200** can be formed is any suitable manner and can be formed from any suitable materials.

[0046] The table frame **100** and the table top **200** can be formed as a single table or from multiple joined tables.

[0047] The table top **200** includes: (1) a first set of upwardly extending spaced apart ramps **240** and **242**; (2) a second set of upwardly extending spaced apart ramps **244** and **246**, at the respective ends of the truss assembly table **10**. The gantry **800** is configured to travel upwardly on each set of ramps and thus be spaced further from the top surface **230** of the table top **200**. This better enables a built truss to move under the gantry **800** as the built truss is ejected from and end of the truss assembly table **10**. The ramps can be only at one end of the truss assembly table and can be otherwise configured. In various embodiments, the truss assembly table includes only one set of ramps that enable the gantry to be moved upwardly. In various other embodiments, present disclosure contemplates that other mechanisms can be employed to raise the gantry if needed.

[0048] In various other embodiments, the table top **200** includes one but not both sets of the upwardly extending spaced apart ramps, at one of the ends of the truss assembly table **10**.

[0049] The multiple evenly spaced-apart truss ejection assemblies **300**, **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, and **300***f* are all identical in this example embodiment, and thus for brevity, only the truss ejection assembly **300** is described herein in detail. The truss ejection assemblies **300**, **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, and **300***f* are configured to simultaneously be in: (1) retracted positions where the truss ejection assemblies are at or below the top surface **230** of the table top **200** of the truss assembly table **10**; and (2) extended positions where the rollers (not all labeled) of the truss ejections assemblies **300**, **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, and **300***f* engage, raise, and eject a truss built on the truss assembly table **10** from an end of the truss assembly table **10**. The quantity and positions of the truss ejections assemblies can vary in accordance with the present disclosure. The truss ejections assemblies are configured to support the entire weight of the built truss in accordance with the present disclosure.

[0050] As best shown in FIGS. **3**, **4**, **5**, and **6**, the example truss ejection assembly **300** generally includes: (1) a support assembly **304**; (2) a truss mover **310**; (3) an actuation assembly **320**; and (4) an actuator **380**.

[0051] The support assembly **304** include supports **305** and **306** and fasteners (not labeled). The support assembly **304** is configured to be fixedly connected to and supported by the table frame **100**. The support assembly **304** is configured to support the truss mover **310**, the actuation assembly **320**, and the actuator **380** such as shown in FIGS. **3**, **4**, **5**, and **6**. The support assembly **304** is made from suitable metal components but can be made from other suitable materials and formed in other suitable configurations.

[0052] The truss mover **310** includes a powered roller such as a cylindrical driven rotatable roller **312** rotatably supported at opposite ends (not labeled) by upwardly extending roller supports **334** and **335** that are part of the actuation assembly **330** (as described below). The truss mover **310** includes a suitable roller actuator (not shown) such as a motor (not shown) controlled by the control system and configured to rotate the roller **312** in either direction. The roller **312** extends transversely to the table top **130**. The truss mover **310** is configured to be moved upwardly through a transversely extending opening (not labeled) in the table top **230** from a retracted position below the table top **230** (such as shown in FIG. **3**) to an extended position at least partially above the table top **230** (such as shown in FIGS. **4** and **5**). In the retracted position, the top of the roller **312** of the truss mover **310** is below or even with the top surface **230** of the table top **200** so as not to interfere with the building of the truss on the second side **10**B of the truss assembly table **10**. In the extended position, the roller **312** of the truss mover **310** (in conjunction with the rollers of the other truss ejection assemblies **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, and **300***f*) is configured to vertically lift and longitudinally move a built truss positioned on the second side **10**B of the truss assembly table **10** as described herein. It should be appreciated that in other embodiments, the truss assembly table include only the features of side **10**B of this example embodiment. In certain such embodiments, the gantry can be narrower.

[0053] The actuation assembly **320** includes: (1) an upper section **330**; (2) a lower section **340**; (3) a first hinge **350** connected to the upper section **330** and to the lower section **340**; (4) a second hinge **360** connected to the upper section **330** and to the lower section **340**; and (5) a first stop **370**. The first hinge **350** connects the upper section **330** to the lower section **340**. The second hinge **360** also connects the upper section **330** to the lower section **340**. The actuation assembly **320** is configured to be moved and particularly pivot upwardly via such hinges **350** and **360** to move the roller **312** from the lower position (such as shown in FIGS. **3** and **4**) to the upper position (such as shown in FIGS. **5** and **6**).

[0054] More specifically, the upper section **330** includes: (1) a roller support **331**; (2) a first pair of legs **332** fixedly connected to and extending downwardly from the roller support **331**; (3) a second pair of legs fixedly **333** connected to and extending downwardly from the roller support **331**; (4) a first roller end support **334** fixedly connected to and extending upwardly from the roller support **331**; and (5) a second roller end support **335** fixedly connected to and extending upwardly from the roller support **331**.

[0055] Each of the first pair of legs **332** is pivotally connected to the first hinge **350** and each of the second pair of legs **333** is pivotally connected to the second hinge **360** such that the pivoting of the hinges **350** and **360** cause these pairs of legs **332** and **333**, the roller support **331**, and the roller **312** to move upwardly or downwardly.

[0056] The roller support **331** extends under the roller **312** and supports the roller **312**. The first roller end support **334** supports a first end of the roller **312** such that the roller **312** can rotate relative to the first roller end support **334**. The second roller end support **335** supports an opposite second end of the roller **312** such that the roller **312** can rotate relative to the second roller end support **335**.

[0057] The lower section **340** includes: (1) an actuation arm **341**; (2) a first actuation hand **342** fixedly connected to and extending upwardly from a first end (not labeled) of the actuation arm **341**; and (3) a second actuation hand **343** fixedly connected to and extending upwardly from an opposite second end (not labeled) of the actuation arm **341**.

[0058] The first actuation hand **342** is pivotally connected to the first hinge **350** and the second actuation hand **343** is pivotally connected to the second hinge **360** such that movement of the actuation arm **342** upwardly or downwardly causes the pivoting of the hinges **350** and **360** upwardly or downwardly.

[0059] The first hinge **350** includes: (1) a first end (not labeled) pivotally connected to the first pair of legs **332** by a first pivot pin (not labeled); (2) a second end (not labeled) pivotally connected to the first actuation hand **342** by a second pivot pin (not labeled); and (3) an intermediate section (not labeled) pivotally mounted on a pivot pin **354** that is fixedly connected to the support assembly **304**. The pivot pin **354** is connected to and supported by supports **305** and **306**.

[0060] The second hinge **360** includes: (1) a first end (not labeled) pivotally connected to the second pair of legs **333** by a third pivot pin (not labeled); (2) a second end (not labeled) pivotally connected to the second actuation hand **343** by a fourth pivot pin (not labeled); and (3) an intermediate section (not labeled) pivotally mounted on a pivot pin **364** that is fixedly connected to the support assembly **304**. The pivot pin **364** is connected to and supported by supports **305** and **306**.

[0061] The first stop **370** is configured to limit the downward movement of the lower section **340** and thus limit the upward movement of the upper section **330** and the roller **312**, as best shown in FIGS. **5** and **6**.

[0062] The actuator **380** includes an upper end (not labeled) and an opposite lower end (not labeled). The lower end of the actuator **380** is pivotally connected to a mount **390** that is fixedly connected to the support assembly **304**. The actuator **380** extends between the lower section **340** of the actuation assembly **320** and the mount **390**. In the example embodiment, the actuator **380** includes a pneumatically powered cylinder (not labeled) controlled by the control system (such as via an air compressor (not shown) controlled by the control system. More specifically, the actuator **380** includes a housing **382** and a piston rod **384** moveable with respect to the housing **382**. The piston rod **384** has a first end (not shown) positioned in the housing **382** and a second end (not labeled) exterior to the housing **382** and pivotally attached by a coupler **388** to the lower section **340** of the actuation assembly **320**. The actuator **380** is pneumatically powered in this example embodiment and thus includes one or more air inlet/outlet ports (not shown). However, it should be appreciated that the actuator **380** can be powered in other suitable manners in accordance with the present disclosure. For example, the actuator can include a suitably electrically powered solenoid.

[0063] The actuator **380** is configured to move the actuation assembly **320**. More specifically, the actuator **380** has a non-actuated configuration shown in FIGS. **3** and **4** and an actuated configuration shown in FIGS. **5** and **6**. In the non-actuated configuration, the actuator **380** causes the actuation assembly **320** to be in its lower position as shown in FIGS. **3** and **4**. In the actuated configuration, the actuator **380** causes the actuation assembly **320** to be in its upper position as shown in FIGS. **5** and **6**. More specifically, when the actuator **380** retracts the piston rod **384**: (1) the piston rod **384** via the coupler **388** pulls the lower section **340** of the actuation assembly **320** downwardly, (2) which causes the hinges **350** and **360** to pivot from their horizontally extending positions shown in FIGS. **3** and **4** to upwardly angled positions shown in FIGS. **5** and **6**, (3) which causes the upper section **310** to move upwardly, and (4) which causes the roller **312** to move upwardly. As the roller **312** moves upwardly, it moves above the top surface **230** of the table top **200** and engages the built truss (not shown) on the table top **200**. This causes the upward movement of the built truss. The roller **312** can then rotate to move the built truss from an end of the truss assembly table **10**.

[0064] Thus, in this example embodiment, the single actuator **380** is configured to move the roller **312** from its retracted position to its extended position to engage and eject a built truss on the second side **10**B of the truss assembly table **10**. The roller **312** co-acts with all of the other rollers of the other truss ejection assemblies **300***a*, **300***b*, **300***c*, **300***d*, **300***e*, and **300***f* to lift and eject the built truss.

[0065] The truss assembly table **10** can have one or two truss end stop assemblies, wherein each truss end stop assembly is positioned toward a respective end of table top **200**. The truss end stop assemblies can be identical so only truss end stop assembly **600** is described herein for brevity.

[0066] The example truss end stop assembly **600** generally includes: (1) a mounting support **602** fixedly connected to and supported by the table frame **100**; (2) a platform **620** vertically movably connected to the mounting support **602**; (3) a platform actuation assembly **640** connected to the mounting support **602** and the platform **620**; and (4) an end stop **670** positioned on, connected to, and supported by the platform **620**. The platform **620** and the end stop **670** are thus supported by the mounting support **602** and movable by the platform actuation assembly **640**: (1) to an engagement position (shown in FIGS. **9** and **10**) where the end stop **670** is configured to engage one or more end truss member(s) of the truss during building of the truss on the second side **10**B of the truss assembly table **10**; and (2) to a non-engagement position (shown in FIGS. **7**, **8**, **11**, and **12**) where the truss end stop assembly **600** is configured to be dis-engaged from the end truss member(s) of the built truss on the second side **10**B of the truss assembly table **10** and enable the built truss to move above the truss end stop assembly **600** as the built truss is ejected from an end of the truss assembly table **10**.

[0067] More specifically, the mounting support **602** includes: (1) a mounting plate **604**; (2) a first guide rail **606** fixedly connected to and supported by the mounting plate **604**; and (3) a second guide rail **608** fixedly connected to and supported by the mounting plate **604**. The mounting support **602** is made from suitable metal components but can be made from other suitable materials and in other suitable configurations.

[0068] The mounting plate **604** is fixedly connected to and supported by the table frame **100**. The mounting plate **604** is configured to support the platform **620**, the platform actuation assembly **640**, and the end stop **670**, such as shown in FIGS. **7**, **8**, **9**, and **11**.

[0069] The first guide rail **606** is fixedly connected to and supported by the mounting plate **604**, extends vertically along the mounting plate **604**, and extends transversely from the mounting plate **604**. The first guide rail **606** is configured to be coupled to the first support bracket **628** of the platform **628** such as described below.

[0070] The second guide rail **608** is fixedly connected to and supported by the mounting plate **604**, extends vertically along the mounting plate **604**, and extends transversely from the mounting plate **604**. The second guide rail **608** is spaced-apart from the first guide rail **606**. The second guide rail **608** is configured to be coupled to the second support bracket **632** of the platform **628** such as described below.

[0071] The platform **620** includes: (1) an upper base **622**; (2) a first support bracket **628**; and (3) a second support bracket **632**. The platform **620** is made from suitable metal components but can be made from other suitable materials and in other suitable configurations.

[0072] The upper base **622** is configured to support the end stop **670**. The upper base **622** includes a top surface **622***a* that is configured to be flush with the top surface **230** of the table top **200** when the end stop assembly **600** (including the platform **620** and the end stop **670**) are in the engagement position (shown in FIGS. **9** and **10**). In such position, the end stop **670** is configured to engage one or more end truss members of the truss during building of the truss on the second side **10**B of the truss assembly table **10**. In such position, parts of the truss members including one or more end members that form the truss can rest on part of the top surface **622***a* of the upper base **622** of the platform **620**.

[0073] The first support bracket **628** includes: (1) an upper portion **629** that is fixedly connected to the bottom of the base **622**; and (2) a side portion **630** connected to a first guide **631** that is coupled to and slidable relative to the first guide rail **606** such that the first bracket **628** is slidable relative to the first guide rail **606** and the mounting plate **604**.

[0074] The second support bracket **632** includes: (1) an upper portion **633** that is fixedly connected to the bottom of the base **622**; and (2) a side portion **634** connected to a second guide **635** that is coupled to and slidable relative to the second guide rail **608** such that the second bracket **632** is slidable relative to the second guide rail **608** and the mounting plate **604**.

[0075] The platform actuation assembly **640** includes: (1) an actuator **643**; (2) a first mount **650**; and (3) a second mount (not shown). The platform actuation assembly **640** is mainly made from suitable metal components but can be made from other suitable materials and in other suitable configurations.

[0076] The actuator **643** includes an upper end (not labeled) and an opposite lower end (not labeled). The lower end (not labeled) of the actuator **643** is pivotally connected to the first mount **650** that is fixedly connected to the mounting plate **604** of the mounting support **602**. The upper end (not labeled) of the actuator **643** is pivotally connected to the second mount that is fixedly connected to the bottom of the base **622** of the platform **620**.

[0077] In the example embodiment, the actuator **643** includes a pneumatically powered cylinder (not shown or labeled) controlled by the control system (such as via an air compressor (not shown) controlled by the control system). The actuator **643** includes a housing (not labeled) and a piston rod **641** moveable with respect to the housing. The piston rod **641** has a first end (not shown) positioned in the housing and a second end (not labeled) exterior to the housing and pivotally attached to the second mount. The actuator **643** is pneumatically powered in this example embodiment and thus includes one or more air inlet/outlet ports (not shown). However, it should be appreciated that the actuator **643** can be powered in other suitable manners in accordance with the present disclosure. For example, the actuator can include a suitably electrically powered solenoid.

[0078] The actuator **643** is configured to move the platform **620** (and thus the end stop **670** on the platform **620**). The actuator **380** has a non-actuated configuration and an actuated configuration. In the non-actuated configuration, the actuator **643** causes the platform **620** to be in its lower position as shown in FIGS. **7**, **8**, **11**, and **12**. In the actuated configuration, the actuator **643** causes the platform **620** to be in its upper position as shown in FIGS. **9** and **10**. Thus, in this example embodiment, the actuator **643** is configured to move the platform **620** (and thus the end stop **670** on the platform **620**) from the upper engagement position to the lower non-engagement position such that a built truss can be moved over the end stop **670** and the platform **620** to be removed from and end of the truss assembly table **10**.

[0079] The end stop **670** includes: (1) an end stop housing **672**; and (2) an extendable stopping member **680** supported by the end stop housing **672**. The end stop **670** is mainly made from suitable metal components but can be made from other suitable materials and in other suitable configurations.

[0080] The end stop housing **672** is mounted on, fixedly connected to, and supported by the upper base **622** of the platform **620**.

[0081] The extendable stopping member **680** is extendable (such as manually slidable) from the end stop housing **672** from a retracted position to an extended position. In the extended position, the extendable stopping member **680** is configured to engage and apply pressure to one or more end truss members of a truss being built on the second side **10**B of the truss assembly table **10**.

[0082] In other embodiments, the end stop **670** can include an end stop actuator is inside the housing **672** and configured to cause the extendable stopping member **680** to move from the retracted position to the extended position and back to the retracted position. Such an end stop actuator can be suitably connected to the housing and also suitably connected to the extendable stopping member. Such an end stop actuator can be configured in any suitable manner and can be controlled by the control system.

[0083] It should thus be appreciated that the end stop assembly **600** is configured to be is an upper position when needed to engage the truss members of the truss being built on the truss assembly table and in a lower position so the built truss can be moved over the end stop assembly **600** to remove the built truss from an end of the truss assembly table.

[0084] The gantry **800** is longitudinally moveable relative to the table top **100** from one end of the table top **100** (as shown in FIGS. **1** and **2**) to an opposite end of the table top **100**. As described above, in this example embodiment, at each end of the table top **100**, the gantry **800** moves upwardly along the respective set of ramps (i.e., either ramps **240** and **242** or ramps **244** and **246**) to make room for a built truss to be moved under the gantry and from the end of the table top **100**. The gantry **800** configured to secure connection plates to the truss members (such as chord members and web members) of a truss being built on the table top **100**. In alternative embodiments, the ramps can be only at one end of the truss assembly table.

[0085] This example gantry **800** includes: (1) a rotatable roller **810**; (2) a housing **830**; (3) a gear assembly **850**; (4) a drive assembly **870**; (5) a drive member **890**; (6) a sensor assembly **900**; and (7) a cover **990**. The gantry **800** can includes other suitable components that are not described herein but known in the truss table assembly industry.

[0086] The housing **830** is supported by and mounted on an A-frame assembly **832** (see FIG. **1**) that partially supports the rotatable roller **810**. The A-frame assembly has a first leg (not labeled) and a second leg (not shown) that ride on rails (not labeled) of the table frame **100** to partially support the roller **810** and the housing **830**. The housing **830** and the A-frame assembly are mainly made from suitable metal components but can be made from other suitable materials and in other suitable configurations. The housing **830** is thus partially supported by and mounted on the rotatable roller **810** such that when the rotatable roller **810** rotates and moves relative to the table top **200**, the housing **830** moves relative to the table top **200**.

[0087] The roller **810** is configured to roll on the table frame **10**. The gantry **800** has additional flanged rollers (not shown) at the bottom of the gantry **800** that run along the bottom rails of the table frame **100** and prevent the gantry **800** from being lifted up when rolling on a truss. The lower rollers are connected to the A-frame **832** with the overall press being adjustable based on the shimming of a main roller bearing (not shown).

[0088] The rotatable roller **810** includes a relatively large cylindrical drum **812**, a first end wall **814** connected to a first end of the drum **812**, and a second end wall (not shown) connected to an opposite second end of the drum **812**. When the roller **810** is rotated, as discussed below, the roller **810** moves over the truss members of the truss being built on the truss assembly table **10** and forces the connection plates into those truss members to securely attach the connection plates to those truss members and thus to securely attach the truss members to each other. The rotatable roller is made from suitable metal components but can be made from other suitable materials and in other suitable configurations.

[0089] The gear assembly **850** includes a first toothed driven gear **852** and a second toothed driven gear **854** each mounted on an axle **853** that is fixedly connected to the end wall **814** of the rotatable roller **810**. The first toothed driven gear **852** and the second toothed driven gear **854** are identical in this example embodiment and are fixedly connected to each other via the main roller axle **853**. When the first toothed driven gear **852** and the second toothed driven gear **854** are rotated in a first rotational direction, they cause the roller **810** to rotate in that first rotational direction and thus cause the gantry **800** to move in a first longitudinal direction relative to the table top **200**. When the first toothed driven gear **852** and the second toothed driven gear **854** are rotated in an opposite second rotational direction, they cause the roller **810** to rotate in that opposite second rotational direction and thus cause the gantry **800** to move in an opposite second longitudinal direction relative to the table top **200**. The gear assembly **850** is made from suitable metal components but can be made from other suitable materials and in other suitable configurations.

[0090] The drive assembly **870** includes: (1) an actuator such as a motor (not shown) and a gearbox (not show); (2) a drive shaft **872** extending from the gearbox; (3) a first drive gear **876** fixedly connected to a drive shaft **872** extending from the gearbox; and (4) a second drive gear **878** fixedly connected to the drive shaft **872** extending from the gearbox. The first drive gear **876** and the second drive gear **878** are identical in this example embodiment and are each fixedly connected to each other via the drive shaft **872**. When the first drive gear **876** and the second drive gear **878** are rotated in the first rotational direction, they cause the drive member **890** to rotate in that first rotational direction and thus cause the driven gears **852** and **854** to rotate in that first rotational direction (which as described above causes the roller **810** to rotate in that first rotational direction and the gantry **810** to move in the first longitudinal direction). When the first drive gear **876** and the second drive gear **878** are rotated in the opposite second rotational direction, they cause the drive member **890** to rotate in that second rotational direction and thus cause the driven gears **852** and **854** to rotate in that second rotational direction (which as described above causes the roller **810** to rotate in that second rotational direction and the gantry **810** to move in the second longitudinal direction). The drive assembly **870** is mainly made from suitable metal components but can be made from other suitable materials and in other suitable configurations.

[0091] The drive member **890** includes a continuous chain journaled about the first and second drive gears **876** and **878** at one end and the first and second tooth driven gears **852** and **854** at the other end. The drive member **890** is made from suitable metal components but can be made from other suitable materials and in other suitable configurations.

[0092] The sensor assembly **900** includes: (1) a support bracket **910**; (2) a sensor **930**; and (3) an electrical connector (not shown).

[0093] The support bracket **910** includes a first member **912** connected to the side wall (not labeled) of the housing **830** and a second member **914** extending transversely outwardly from the side wall of the housing **830**. The support bracket **910** is formed from metal in this example embodiment but can be made from other materials and configured in other manners. The support bracket **910** is positioned relatively close to the driven gears **852** and **854**, such as best shown in FIGS. **13**, **14**, **16**, and **17**.

[0094] The second member **914** of the support bracket **910** supports the sensor **930** in a location and position aligned with the gear assembly **850** and specifically aligned with the outer second driven gear **854**. In this example embodiment, the sensor **930** is aligned with driven gear **854** such that the sensor **930** can send a signal in this example embodiment when each peak such as peak **854**P (see FIG. **17**) of the gear **854** passes the sensor **930**, or in alternative embodiments when each trough such as trough **854**T (see FIG. **17**) of the gear **854** passes the sensor **930**. Thus, as the driven gear **854** is rotated in either rotational direction, the sensor **930** will send a series of signals to the control system. The signals will be based on the speed of the rotation of the driven gear **845** and thus the speed of rotation of the roller **810** (because the driven gear **854** is fixedly connected to the roller **810**). The speed is thus directly related to the distance of movement of the roller **810** and the gantry **800**.

[0095] The sensor **930** is an inductive sensor in this example embodiment but can be any suitable type of sensor in other embodiments. The sensor **930** creates a magnetic field and when each peak of the gear **854** passes through the magnetic field, the peak in the magnetic field is sensed by the sensor **930**. The sensor **930** is configured to send electrical signals to the control system based on such sense peaks. In this example embodiment, the sensor assembly **930** includes one sensor, but it should be appreciated that the sensor assembly can include multiple sensors. In various such embodiments, such multiple sensors can be positioned adjacent to each other and be respectively associated with the different driven gears. In various other embodiments, the sensor assembly can include multiple sensors associated with the same driven gear to better ensure accurate signals sent to the control system.

[0096] In various embodiments, the control system is configured to use such signals from the sensor **930** to determine the exact amount of movement of the roller **810** and of the gantry **800** relative to the table top **100**. More specifically, the control system can be provided with information regarding a position of the roller **810** and of the gantry **800** relative to the table top **100** such as an initial position at either of the first end of the table top **200** such as shown in FIGS. **1** and **2**, or at a second end of the table top **200**. When the motor of the drive assembly **870** rotates the drive shaft **872** of the motor which causes the first and second drive gears **876** and **878** to rotate, which causes the drive member **890** to rotate, which causes the first and second driven gears **852** and **854** to rotate, the sensor **930** sends signals to the control system that represent the amount of rotation of the driven gear **854** (based on quantity of the peaks or troughs) that pass the sensor **930**. The control system can then use this information to determine the exact amount of longitudinal distance the roller **810** and thus the gantry **800** have moved relative to the table top **200**.

[0097] In various embodiments, the control system is configured to use such signals received from the sensor **930** to control the exact amount of movement of the roller **810** and thus of the gantry **800** relative to the table top **100**. More specifically, the control system can, based on information regarding an initial position of the roller **810** and of the gantry **800** relative to the table top **100** (such as a position at either of the first end of the table top **200** such as shown in FIGS. **1** and **2** or at a second end of the table top **200**), control the amount of movement of the roller **810** and thus of the gantry **800** relative to the table top **100**. In other words, if the control system wants to move the roller **810** and the gantry **800** a determined distance relative to the table top **200**, the control system can cause the motor of the drive assembly **870** to rotate the drive shaft **872** of the motor to cause the first and second drive gears **876** and **878** to rotate, to cause the drive member **890** to rotate, to cause the first and second driven gears **852** and **854** to rotate. The sensor **930** sends signals regarding the movement of the driven gear **854** to the control system that represent the amount of rotation of the driven gears (based on quantity of the peaks or troughs of the driven gear **854**) that pass the sensor **930**. The control system uses this information to control the exact amount of longitudinal distance that the roller **810** and the gantry **800** move relative to the table top **200**.

[0098] In various embodiments, the control system (not shown) includes one or more manually controlled switching mechanisms.

[0099] In various embodiments, the control system includes one or more PLC boards or is integrated into one or more PLC boards.

[0100] In various embodiments, the control system includes one or more processing devices communicatively connected to one or more memory devices.

[0101] In various embodiments, the control system includes a programmable logic control system. The processing device can include any suitable processing device such as, but not limited to, a general-purpose processor, a special-purpose processor, a digital-signal processor, one or more microprocessors, one or more microprocessors in association with a digital-signal processor core, one or more application-specific integrated circuits, one or more field-programmable gate array circuits, one or more integrated circuits, and/or a state machine.

[0102] In various embodiments, the control system includes one or more memory devices. Each memory device can include any suitable memory device such as, but not limited to, read-only memory, random-access memory, one or more digital registers, cache memory, one or more semiconductor memory devices, magnetic media such as integrated hard disks and/or removable memory, magneto-optical media, and/or optical media. Each memory device stores instructions executable by the processing device to control operation of the truss assembly table **10**.

[0103] In various embodiments, the control system is communicatively and operably connected to the actuators including the servo motors, sensors, the operator interface, and the power supply, and configured to receive signals from and send signals to those components.

[0104] In various embodiments, the control system is communicatively connectable (such as via Wi-Fi, Bluetooth, near-field communication, or other suitable wireless communications protocol) to an external device, such as a computing device, to send information to and receive information from that external device.

[0105] The operator interface can include a suitable display screen with a touch panel. In such embodiments with a display screen, the display screen is configured to display information regarding the truss assembly table **10**, and the touch screen is configured to receive operator inputs. The operator interface is communicatively connected to the control system to send signals to the control system and to receive signals from the control system. Other embodiments of the truss assembly table **10** do not include a touch panel. Still other embodiments of the truss assembly table **10** do not include a display assembly. Certain embodiments of the truss assembly table **10** include a separate pushbutton panel instead of a touch panel beneath or integrated with the display screen. In certain embodiments of the truss assembly table **10**, the operator interface includes one or more pushbuttons (and associated light) and no display screen or touch panel.

[0106] In various embodiments, the power supply is electrically connected to (via suitable wiring and other components) and configured to power several components of the truss assembly table **10**. In various embodiments, the power supply can include a pneumatic air power supply.

[0107] It will be understood that modifications and variations may be affected without departing from the scope of the novel concepts of the present invention, and it is understood that this application is to be limited only by the scope of the claims.

1. A truss assembly table comprising:
a table frame;
a table top supported by the table frame; and
a plurality of spaced-apart truss ejection assemblies supported by the table frame, each truss ejection assembly including:
a support assembly supported by the table frame,
an actuation assembly supported by the support assembly,
an actuator supported by the support assembly, and
a truss mover supported by the support assembly and including a roller, the truss mover movable by the actuator and actuation assembly from a retracted position at least partially below the table top to an extended position at least partially above the table top, wherein in the retracted position, a top of the roller is below or even with a top surface of the table top, and wherein in the extended position, the top of the roller is above the top surface of the table top and configured to vertically lift and longitudinally guide a truss built on the table top.

2. The truss assembly table of claim **1**, wherein the roller is a cylindrical driven rotatable roller rotatably supported at opposite ends by upwardly extending roller supports.

3. The truss assembly table of claim **1**, the actuation assembly includes first and second hinges configured to pivot to move the truss mover from the retracted position to the extended position.

4. The truss assembly table of claim **1**, wherein the actuation assembly includes an upper section, a lower section, a first hinge connected to the upper section and to the lower section, a second hinge connected to the upper section and to the lower section, wherein the first hinge connects the upper section to the lower section, and wherein the second hinge connects the upper section to the lower section.

5. The truss assembly table of claim **4**, wherein the upper section includes a roller support, a first pair of legs fixedly connected to and extending downwardly from the roller support, a second pair of legs fixedly connected to and extending downwardly from the roller support, a first roller end support fixedly connected to and extending upwardly from the roller support, and a second roller end support fixedly connected to and extending upwardly from the roller support.

6. The truss assembly table of claim **5**, wherein each of the first pair of legs is pivotally connected to the first hinge, and each of the second pair of legs is pivotally connected to the second hinge such that the pivoting of the first and second hinges causes these pairs of legs, the roller support, and the roller to move upwardly or downwardly.

7. The truss assembly table of claim **6**, wherein the lower section includes an actuation arm, a first actuation hand fixedly connected to and extending upwardly from a first end of the actuation arm, and a second actuation hand fixedly connected to and extending upwardly from an opposite second end of the actuation arm, wherein the first actuation hand is pivotally connected to the first hinge and the second actuation hand is pivotally connected to the second hinge such that movement of the actuation arm upwardly or downwardly causes the pivoting of the first and second hinges.

8. The truss assembly table of claim **1**, wherein the actuator has a non-actuated configuration and an actuated configuration, wherein in the non-actuated configuration, the actuator causes the actuation assembly to be in a lower position, and in the actuated configuration, the actuator causes the actuation assembly to be in an upper position.

9. A truss ejection assembly for a truss assembly table including a table frame and a table top, the truss ejection assembly comprising:

a support assembly supportable by the table frame;

an actuation assembly supported by the support assembly;

an actuator supported by the support assembly; and

a truss mover supported by the support assembly and including a roller, the truss mover movable by the actuator and actuation assembly from a retracted position to an extended position, wherein in the extended position, the roller is configured to vertically lift and longitudinally guide a truss built on the table top.

10. The truss ejection assembly of claim **9**, wherein the roller is a cylindrical driven rotatable roller rotatably supported at opposite ends by upwardly extending roller supports.

11. The truss ejection assembly of claim **9**, the actuation assembly includes first and second hinges configured to pivot to move the truss mover from the retracted position to the extended position.

12. The truss ejection assembly of claim **9**, wherein the actuation assembly includes an upper section, a lower section, a first hinge connected to the upper section and to the lower section, a second hinge connected to the upper section and to the lower section, wherein the first hinge connects the upper section to the lower section, and wherein the second hinge connects the upper section to the lower section.

13. The truss ejection assembly of claim **12**, wherein the upper section includes a roller support, a first pair of legs fixedly connected to and extending downwardly from the roller support, a second pair of legs fixedly connected to and extending downwardly from the roller support, a first roller end support fixedly connected to and extending upwardly from the roller support, and a second roller end support fixedly connected to and extending upwardly from the roller support.

14. The truss ejection assembly of claim **13**, wherein each of the first pair of legs is pivotally connected to the first hinge, and each of the second pair of legs is pivotally connected to the second hinge such that the pivoting of the first and second hinges causes these pairs of legs, the roller support, and the roller to move upwardly or downwardly.

15. The truss ejection assembly of claim **14**, wherein the lower section includes an actuation arm, a first actuation hand fixedly connected to and extending upwardly from a first end of the actuation arm, and a second actuation hand fixedly connected to and extending upwardly from an opposite second end of the actuation arm, wherein the first actuation hand is pivotally connected to the first hinge and the second actuation hand is pivotally connected to the second hinge such that movement of the actuation arm upwardly or downwardly causes the pivoting of the first and second hinges.

16. The truss ejection assembly of claim **9**, wherein the actuator has a non-actuated configuration and an actuated configuration, wherein in the non-actuated configuration, the actuator causes the actuation assembly to be in a lower position, and in the actuated configuration, the actuator causes the actuation assembly to be in an upper position.

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